

Background Guide

YUGEN 6.0 MUN

Agenda:

Deliberation on safeguarding critical undersea and maritime infrastructure against hybrid threats, with special emphasis on threats to undersea communication cables.

Letter from the Executive Board

Dear Delegates,

We are very pleased to welcome you to the simulation of the **UNGA: DISEC** at **OBUL REDDY MUN 2026**. It is an honour to serve as your Executive Board for the duration of the conference. This Background Guide is designed to give you an insight into the case at hand, so we hope this acts as only a catalyst for furthering your research and not limited to just this guide. Please refer to it carefully. Remember, a thorough understanding of the problem is the first step to solving it.

Do understand that this Background Guide is in no way exhaustive and is only meant to provide you with enough background information to establish a platform for beginning the research. Delegates are highly recommended to do a good amount of research beyond what is covered in the Guide. The guide cannot be used as proof during the committee proceedings under any circumstances.

We understand that MUN conferences can be an overwhelming experience for first-timers but it must be noted that our aspirations from the delegates are not how experienced or articulate they are. Rather, we want to see how one manages the balance to respect disparities and differences of opinion and work around this while extending their foreign policy to present comprehensive solutions without compromising on their self-interests and initiate consensus building.

New ideas are by their very nature disruptive, but far less disruptive than a world set against the backdrop of stereotypes and regional instability due to which reform is essential in policy making and conflict resolution. At any point during your research, do not hesitate to contact the Executive Board Members for clarifications or in case you need help in any other aspect. We look forward to a fruitful discussion and an enriching experience with all of you.

Best regards,

SHYAM D
Chairperson

RITHIK
Vice Chair

Important Points to Remember

A few aspects that delegates should keep in mind while preparing:

1. **Procedure:** The purpose of putting in procedural rules in any committee is to ensure a more organized and efficient debate. The committee will follow the UNA-USA Rules of Procedure. Although the Executive Board shall be fairly strict with the Rules of Procedure, the discussion of the agenda will be the main priority. So, delegates are advised not to restrict their statements due to hesitation regarding procedure.
2. **Foreign Policy:** Following the foreign policy of one's country is the most important aspect of a Model UN Conference. This is what essentially differentiates a Model UN from other debating formats. To violate one's foreign policy without adequate reason is one of the worst mistakes a delegate can make.
3. **Role of the Executive Board:** The Executive Board is appointed to facilitate debate. The committee shall decide the direction and flow of debate. The delegates are the ones who constitute the committee and hence must be uninhibited while presenting their opinions/stance on any issue. However, the Executive Board may put forward questions and/or ask for clarifications at all points of time to further debate and test participants.
4. **Nature of Source/Evidence:** This Background Guide is meant solely for research purposes and must not be cited as evidence to substantiate statements made during the conference. Evidence or proof for substantiating statements made during formal debate is acceptable from the following sources:
 - a. **United Nations:** Documents and findings by the United Nations or any related UN body is held as a credible proof to support a claim or argument. Multilateral Organizations: Documents from international organizations like OIC, NATO, SAARC, BRICS, EU, ASEAN, the International Court of Justice, etc. may also be presented as credible sources of information.
 - b. **Government Reports:** These reports can be used in a similar way as the State Operated News Agencies reports and can, in all circumstances, be denied by another country.
 - c. **News Sources:**
 1. Reuters: Any Reuters article that clearly makes mention of the fact or is in contradiction of the fact being stated by a delegate in council.
 2. State operated News Agencies: These reports can be used in the support of or against the State that owns the News Agency. These reports, if credible or substantial enough, can be used in support of

or against any country as such but in that situation, may be denied by any other country in the council. Some examples are – RIA Novosti (Russian Federation), Xinhua News Agency (People's Republic of China), etc.

****Please Note: Reports from NGOs working with UNESCO, UNICEF and other UN bodies will be accepted. Under no circumstances will sources like Wikipedia, or newspapers like the Guardian, Times of India, etc. be accepted. However, notwithstanding the criteria for acceptance of sources and evidence, delegates are still free to quote/cite from any source as they deem fit as a part of their statements.*

Guidelines

- Read the entirety of the background guide in the order it was written. Make sure to highlight the names of specific treaties, documents, resolutions, conventions, international bodies, events and any other specific incidents so that you can get back to them later and do a lot more thorough research.
- Understand some of the basic details regarding the country that you've been allotted whether this be the capital, current affairs regarding geopolitical situation, political hierarchy etc. While not strictly necessary, you never know when this can turn out to be handy. Geography Now's A - Z Country List has been a particularly helpful resource for this.
- Use a search engine of your choice to create as many tabs as possible for the highlighted terms from your background guide. Wikipedia or a YouTube video act as a great way to get a brief summary of the incidents at hand but such sources (especially Wikipedia articles) cannot be used in committee as sources.
- Delve into deeper research regarding the particular position of your allocation with the agenda at hand. Try searching for the voting stances of your allocation in related conventions and understanding the reasons for voting as so. UN Press Releases are also a helpful source for this matter.
- Find the website for the foreign ministry of the country you have been assigned alongside the "Permanent Mission of COUNTRY to the United Nations" website and search for a key term relating to the agenda, this should often give you statements from recent press conferences or UN committee sessions that can act as valuable sources of information in forming a position.
- Keep a handy copy of the Charter of the United Nations, whether as a .pdf file extension or a physical copy works. This contains the founding principles of the

United Nations and contains articles that lay out the mandate of the six bodies that the United Nations is primarily divided into. Spend some additional time researching the specific mandate and functions of the committee that you have been assigned.

- The Executive Board may ask for the source of a statement that a delegate makes in committee either during a Point of Order circumstance or if said statement stands to be of interest to the Executive Board. Therefore, it is recommended that delegates keep track of their sources when making / disputing a claim and also ensure their validity. Please do remember that while you as a delegate are allowed to cite any source you wish during committee.

Hierarchy of evidence

Evidence can be presented from a wide variety of sources, but not all sources are treated as equal. Here's the hierarchy in which evidence is categorised:

Tier 1: Includes any publication, statement, resolution, or document released by any of the Nations' official organs or committees; any publication, statement, or document released by a UN member state in its own capacity. The evidence falling in this tier is considered most reliable during the simulation.

Tier 2: Includes: any news article published by any official media source that is owned and controlled by a UN member state. E.g.: Xinhua News (China), Prasar Bharti (India), BBC (United Kingdom) etcetera. The evidence falling in this tier is considered sufficiently reliable in case no other evidence from any Tier 1 source is available on that particular fact, event, or situation.

Tier 3: Includes: any publication from news sources of international reputation such as Reuters, The New York Times, Agence-France Presse, etcetera. The evidence falling under this tier is considered the least reliable for the purposes of this simulation. Yet, if no better source is available in a certain scenario, it may be considered.

Foreign Policy and Foreign Relations

Foreign policy, in simple terms, is what your country aims to achieve in regard to the issue at hand or in general with its relations with other countries.

1. What role must foreign policy play in your research?

Understanding the foreign policy of your country must be a checkbox that you tick off at the very beginning of your research.

Your foreign policy should dictate everything from the arguments you make, the reasoning you give for making those arguments, and the actions you take in the Council.

2. Where do I look to find foreign policy?

Most of the time, foreign policy is not explicitly stated. It must be inferred from the actions and statements issued by the country. Reading the meeting records from previous meetings of UNSC (or any other UN body where your country might have spoken on the issue) is a great place to start. If such records are unavailable, look for statements from your country's Foreign Ministry (or equivalent like Ministry of External Affairs, Ministry for Foreign Affairs etcetera) and top leadership (PM, Pres., Secretary of State, Defence Minister).

Foreign Relations on the other hand refers to the diplomatic ties that one country has with another and considers elements such as the mutual presence of embassies, consulates, ambassadors & diplomatic dialogue. More often than not, foreign policy is what will be of your primary concern during the MUN, but it is important to also consider any extremities in your allotted country's foreign relations.

Introduction to the Committee

The United Nations General Assembly (UNGA) is one of the six organs of the United Nations (UN) and the primary policy making and representative organ of the United Nations where all of the 193 members of the UN are represented in the General Assembly. The Disarmament and International Security Committee is the first committee under the General Assembly that deals with disarmament, global challenges and threats to world peace.

The six committees of the United Nations General Assembly being:

- First Committee - Disarmament & International Security,
- Second Committee - Economic & Financial Council,
- Third Committee - Social, Humanitarian & Cultural Issues,
- Fourth Committee - Special Political & Decolonization Committee,
- Fifth Committee - Administrative & Budgetary, &
- Sixth Committee - Legal.

Mandate

The Disarmament and International Security Committee is mandated to address the nuclear, chemical, biological, conventional, etc. weapon proliferation and to deal with the issues threatening international security and peace.

DISEC makes recommendations to the UNGA regarding the resolutions and establishes principles and international cooperation between states to maintain global security. It considers all disarmament and international security matters within the scope of the Charter or relating to the powers and functions of any other organ of the United Nations; the general principles of cooperation in the maintenance of international peace and security, as well as principles governing disarmament and the regulation of armaments; promotion of cooperative arrangements and measures aimed at strengthening stability through lower levels of armaments.

DISEC works closely with other UN bodies and agencies like the United Nations Office on Disarmament Affairs (UNODA) and International Atomic Energy Agency (IAEA). The First Committee sessions are structured into three distinctive stages:

1. General debate,
2. Thematic discussions, &
3. Action on Drafts.

Part I: Introduction

Why Undersea Infrastructure Matters

The modern international system depends not only on visible infrastructure such as roads, ports, airports, and satellites, but also on an extensive network of critical infrastructure located beneath the world's oceans. Collectively referred to as **critical undersea infrastructure (CUI)**, these assets include submarine communication cables, offshore energy installations, pipelines, power interconnectors, military surveillance systems, and scientific monitoring equipment. Although largely hidden from public view, they form the backbone of global connectivity, economic activity, energy security, and strategic defence.

Submarine fibre-optic cables facilitate the overwhelming majority of international data transmission, enabling internet connectivity, financial transactions, diplomatic communications, cloud computing, and military command and control systems. Offshore oil and gas platforms, together with undersea pipelines, contribute significantly to global energy production and transportation, while submarine electricity cables increasingly support cross-border energy integration and renewable energy development. Scientific infrastructure, including ocean observatories and tsunami warning systems, provides critical data for disaster preparedness, climate monitoring, and marine research.

The disruption or destruction of these assets can produce cascading consequences across multiple sectors. A damaged communication cable may interrupt banking systems, healthcare services, aviation, and emergency communications, while attacks on energy infrastructure may result in fuel shortages, economic instability, or environmental disasters. Unlike terrestrial infrastructure, undersea systems are often located in remote and difficult-to-access environments, making surveillance, maintenance, and repair both technically complex and financially demanding.

Consequently, safeguarding undersea infrastructure has evolved from a primarily commercial or engineering concern into a matter of national security and international peace. As states become increasingly interconnected through digital networks and shared energy systems, the resilience of critical undersea infrastructure has become indispensable to global stability.

Global Dependence on Submarine Infrastructure

The global economy is deeply reliant on undersea infrastructure, particularly submarine communication cables. Contrary to popular belief, satellites carry only a small fraction of international communications. Instead, **approximately 95–99% of global intercontinental digital data**—including internet traffic, financial transactions, government communications, cloud services, and military information—travels through an extensive network of more than **600 submarine fibre-optic cables** spanning over **1.5 million kilometres** across the world's oceans.

These cables connect continents, facilitate global commerce, and enable the functioning of multinational corporations, financial markets, academic institutions, and governments. Every international online activity—from video conferencing and e-commerce to stock exchanges and digital banking—relies on the uninterrupted operation of these networks. Major technology companies such as Google, Microsoft, Meta, and Amazon have become significant investors in submarine cable infrastructure, reflecting its growing strategic and commercial importance.

Global dependence extends beyond digital connectivity. Offshore oil and gas platforms account for a substantial proportion of global hydrocarbon production, while submarine pipelines transport energy resources across national boundaries. Additionally, submarine electricity interconnectors, particularly High Voltage Direct Current (HVDC) cables, enable countries to exchange electricity, integrate renewable energy sources, and enhance the resilience of regional power grids.

This increasing reliance means that even localized disruptions can generate widespread international consequences. Damage to a single strategic cable or pipeline may affect multiple countries simultaneously, demonstrating that undersea infrastructure has become a shared global asset whose protection requires international cooperation.

Evolution of Maritime Security

Traditionally, maritime security focused on ensuring the safe movement of ships, protecting sea lanes of communication, combating piracy, preventing smuggling, and maintaining naval superiority during armed conflict. Throughout much of the twentieth century, maritime security was primarily concerned with surface and submarine warfare, freedom of navigation, and the defence of strategic maritime chokepoints.

However, technological advancements and globalization have fundamentally transformed the maritime security landscape. The rapid expansion of submarine communication networks, offshore energy infrastructure, and underwater surveillance systems has shifted attention from the sea surface to the seabed. Oceans are no longer viewed solely as routes for transportation but as locations containing critical infrastructure essential for economic prosperity, national defence, and societal functioning.

The emergence of cyber capabilities, autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), advanced seabed mapping technologies, and sophisticated underwater surveillance systems has further expanded the range of potential threats. Infrastructure that was once considered relatively secure due to its underwater location is now increasingly vulnerable to espionage, sabotage, cyber-physical attacks, and covert operations.

As a result, modern maritime security has evolved into a multidimensional concept encompassing not only naval defence but also the protection of critical undersea infrastructure, maritime domain awareness, cyber resilience, environmental security, and

international cooperation. This evolution reflects the growing recognition that safeguarding the seabed is as important as protecting the waters above it.

Why the Issue Has Become Urgent After 2022

While concerns regarding the vulnerability of undersea infrastructure have existed for decades, the issue gained unprecedented international attention following a series of high-profile incidents beginning in **2022**. These events demonstrated that critical underwater assets could become deliberate targets during periods of geopolitical tension, exposing significant gaps in existing monitoring, attribution, and protection mechanisms.

The most prominent incident was the **Nord Stream pipeline explosions** in September 2022, which damaged major natural gas pipelines in the Baltic Sea. Although investigations continue and responsibility remains contested, the incident underscored the strategic vulnerability of offshore energy infrastructure and illustrated how attacks on undersea assets could produce economic, political, and security repercussions far beyond the immediate area of damage.

Subsequent incidents further heightened international concern. Damage to the **Balticconnector gas pipeline** and multiple submarine communication cables in the Baltic Sea during 2023 prompted investigations into possible sabotage involving commercial vessels. In 2024, several submarine communication cables in the **Red Sea** were severed, disrupting internet connectivity between Europe, Asia, and Africa amid heightened regional instability. Repeated cable disruptions around **Taiwan** have also drawn attention to the strategic use of infrastructure interference in areas of geopolitical competition.

Simultaneously, rapid advances in underwater technologies—including autonomous underwater vehicles, remotely operated systems, precision seabed mapping, and sophisticated cyber capabilities—have increased both the capability to target undersea infrastructure and the difficulty of detecting malicious activity. The growing use of hybrid tactics, in which states or proxy actors employ sabotage, cyber operations, disinformation, and economic pressure below the threshold of armed conflict, has further complicated responses under existing international legal and security frameworks.

These developments have transformed the protection of critical undersea and maritime infrastructure from a specialized technical concern into a central issue of international peace and security. Today, governments, international organizations, and private infrastructure operators increasingly recognize that ensuring the resilience of undersea infrastructure is essential not only for safeguarding national interests but also for maintaining the stability and functioning of the global international system.

2. Understanding Hybrid Threats

The security environment of the twenty-first century has undergone a profound transformation. Traditionally, armed conflicts were characterized by conventional military engagements between states, involving clearly identifiable combatants and battlefields. However, technological advancements, globalization, and increasing geopolitical competition have led to the emergence of more complex forms of conflict that often blur the distinction between peace and war. Rather than relying solely on military force, states and non-state actors increasingly employ a combination of political, economic, cyber, informational, and covert methods to influence or weaken their adversaries. Collectively, these methods are referred to as **hybrid threats**.

Hybrid threats are deliberately designed to exploit legal ambiguities, create uncertainty regarding attribution, and achieve strategic objectives while avoiding the political and military consequences associated with open armed conflict. In the context of critical undersea and maritime infrastructure, such threats have become particularly significant due to the strategic importance, geographical isolation, and relative vulnerability of undersea assets.

Hybrid Warfare

Hybrid warfare refers to the coordinated use of conventional military capabilities alongside non-military instruments to achieve political or strategic objectives. Unlike traditional warfare, hybrid warfare integrates multiple domains—including military operations, cyberattacks, economic pressure, information campaigns, intelligence activities, proxy forces, sabotage, and diplomatic coercion—into a unified strategy.

The defining characteristic of hybrid warfare is not the use of any single method but the simultaneous employment of diverse tools that reinforce one another. An actor may launch cyberattacks against communication networks, spread disinformation through social media, impose economic restrictions, support proxy groups, and conduct covert sabotage of critical infrastructure while avoiding direct military confrontation. Such an approach complicates attribution, delays decision-making, and reduces the likelihood of a conventional military response.

For critical undersea infrastructure, hybrid warfare may involve disrupting submarine communication cables, interfering with offshore energy infrastructure, manipulating maritime navigation systems, or conducting cyber operations against infrastructure management systems, often while maintaining plausible deniability.

Grey-Zone Conflict

Grey-zone conflict refers to competitive actions undertaken between the conditions of peace and declared war. These activities deliberately remain below the threshold that would generally justify a military response under international law or trigger collective defence mechanisms such as those contained within military alliances.

Grey-zone operations exploit uncertainty by employing methods that are difficult to classify as acts of war. These may include maritime harassment, cyber intrusions, infrastructure interference, election interference, economic pressure, intelligence operations, or the deployment of civilian vessels for strategic purposes.

In maritime environments, grey-zone tactics frequently involve coast guards, fishing fleets, research vessels, or commercial ships conducting activities that support national strategic objectives while avoiding direct military engagement. Such operations allow states to gradually alter the status quo without provoking large-scale escalation.

Coercion Below the Threshold of War

Coercion below the threshold of war refers to actions intended to compel another state to alter its behaviour without resorting to open armed conflict. Rather than employing direct military force, an actor seeks to impose costs or create pressure through methods that remain legally or politically ambiguous.

Such coercive measures may include cyberattacks, economic sanctions, interference with critical infrastructure, diplomatic isolation, maritime blockades, or covert sabotage. The objective is to influence political decision-making while avoiding actions that would clearly constitute an armed attack under international law.

Disrupting undersea communication cables, interfering with offshore energy supplies, or targeting maritime logistics networks can serve as effective coercive tools by imposing significant economic and political costs without necessarily triggering military retaliation.

Information Warfare

Information warfare involves the deliberate use, manipulation, disruption, or control of information to influence public opinion, government decision-making, military operations, or societal behaviour. Unlike cyber warfare, which primarily targets digital systems, information warfare focuses on shaping perceptions and controlling narratives.

Techniques commonly associated with information warfare include propaganda, disinformation, misinformation, psychological operations, manipulated media, social media influence campaigns, and strategic communication. Modern developments in artificial intelligence have further increased the sophistication of information operations through the creation of convincing deepfakes, automated influence campaigns, and synthetic media.

Information warfare often accompanies physical or cyber attacks. For example, following damage to submarine infrastructure, competing actors may circulate conflicting narratives regarding responsibility, thereby complicating investigations and delaying international responses.

Economic Coercion

Economic coercion refers to the use of economic measures to pressure another state into changing its political or strategic behaviour. Instead of employing military force, states may exploit trade relationships, financial systems, technology exports, investment flows, or energy dependence to achieve foreign policy objectives.

Examples include economic sanctions, export controls, import restrictions, asset freezes, investment limitations, and manipulation of energy supplies. In some instances, disruption of critical maritime infrastructure may itself function as a form of economic coercion by interrupting trade routes, communication networks, or energy transportation.

As global economic interdependence increases, economic coercion has become an increasingly prominent component of hybrid strategies.

Cyber Warfare

Cyber warfare involves offensive and defensive operations conducted through cyberspace to disrupt, degrade, manipulate, or destroy digital systems and networks. Targets often include government institutions, military systems, financial networks, telecommunications infrastructure, transportation systems, and critical infrastructure.

Within the maritime domain, cyber operations may target port management systems, vessel navigation software, offshore energy control systems, submarine cable management networks, or satellite communications. Successful cyberattacks may disrupt operations without causing visible physical destruction, making detection and attribution particularly difficult.

Cyber warfare frequently complements physical sabotage, allowing attackers to maximize disruption while complicating recovery efforts.

Proxy Actors

Proxy actors are individuals, organizations, or armed groups that conduct operations on behalf of another state while allowing that state to deny direct responsibility. States frequently employ proxy actors to pursue strategic objectives while minimizing diplomatic consequences and reducing the risk of military escalation.

Proxy actors may include private military companies, cybercriminal organizations, separatist groups, insurgent movements, intelligence-linked organizations, or politically aligned militias. In cyberspace, governments may sponsor hacker groups to conduct espionage or infrastructure attacks while maintaining plausible deniability.

The involvement of proxy actors significantly complicates attribution and legal accountability under international law, making effective international responses more challenging.

State-Sponsored Sabotage

State-sponsored sabotage refers to deliberate acts of destruction or disruption directed against infrastructure, facilities, or equipment with the support, authorization, or coordination of a government. Such operations are generally covert and intended to weaken an adversary's capabilities without initiating open military conflict.

Targets may include submarine communication cables, pipelines, electricity interconnectors, offshore energy platforms, transportation systems, or industrial facilities. These operations often involve specialized military units, intelligence agencies, autonomous underwater vehicles, or proxy actors.

Because state-sponsored sabotage is usually designed to conceal the identity of those responsible, attribution remains one of the greatest challenges facing the international community.

Maritime Militias

Maritime militias consist of civilian fishing vessels, commercial ships, or other maritime actors that receive direction, coordination, or support from governments to advance strategic or political objectives. Although these vessels generally appear civilian, they may conduct surveillance, intelligence collection, obstruction of foreign vessels, assertion of territorial claims, or logistical support for military operations.

Maritime militias are particularly significant in contested maritime regions where governments seek to strengthen territorial claims without deploying regular naval forces. Their civilian appearance complicates legal classification under international humanitarian law and creates challenges for escalation management.

The use of maritime militias represents a prominent example of grey-zone tactics because they deliberately blur the distinction between civilian and military actors.

Private Actors

Private actors play an increasingly significant role in the ownership, operation, and maintenance of critical undersea infrastructure. Telecommunications companies, technology firms, energy corporations, cable consortia, engineering contractors, cybersecurity providers, and maritime operators collectively own or manage much of the world's submarine infrastructure.

Consequently, the security of undersea infrastructure cannot be ensured solely through governmental action. Effective protection requires close cooperation between states and private entities in areas such as cybersecurity, information sharing, infrastructure monitoring, emergency response, and repair capabilities.

The growing involvement of private actors also raises important legal and governance questions concerning responsibility, liability, information security, and the protection of commercially owned infrastructure during periods of international tension.

Distinguishing Different Forms of Attack

Understanding the differences between various forms of hostile activity is essential for assessing threats to maritime infrastructure and determining appropriate legal and political responses.

Conventional Attack

A conventional attack involves the overt use of military force by one state against another during an armed conflict. Such attacks typically employ regular armed forces—including naval fleets, aircraft, missiles, or ground forces—and are generally governed by the principles of international humanitarian law. Responsibility for the attack is usually clear, and the objective is often to achieve military or territorial advantage.

Terrorism

Terrorism is the intentional use or threat of violence against civilians or civilian infrastructure to create fear and compel governments or societies to achieve political, ideological, or religious objectives. Terrorist organizations often seek psychological impact beyond the immediate physical damage, using violence to influence public behaviour or government policy.

Piracy

Piracy consists of illegal acts of violence, detention, or robbery committed at sea by private individuals for personal or financial gain. Under international law, piracy typically occurs on the high seas and is motivated by economic benefit rather than political objectives. Modern piracy remains concentrated in regions with weak maritime governance, such as parts of the Gulf of Guinea and, historically, the Horn of Africa.

Sabotage

Sabotage refers to the deliberate damage, destruction, or interference with infrastructure, equipment, or systems to reduce their effectiveness or disrupt normal operations. Sabotage may be conducted by states, proxy actors, insiders, criminal organizations, or individuals and may occur during both peacetime and armed conflict. Unlike terrorism or piracy, sabotage primarily seeks operational disruption rather than financial gain or psychological intimidation.

Hybrid Attack

A hybrid attack combines multiple conventional and non-conventional methods into a coordinated campaign intended to achieve strategic objectives while remaining below the threshold of full-scale war. Such attacks frequently integrate cyber operations, economic pressure, information warfare, proxy actors, diplomatic coercion, and covert sabotage.

For example, a state may simultaneously launch cyberattacks against port systems, damage submarine communication cables through covert means, spread disinformation regarding the incident, and employ proxy actors to deny involvement. Individually, each action may appear limited; collectively, however, they can significantly weaken an adversary while avoiding direct military confrontation. This integrated and ambiguous nature distinguishes hybrid attacks from conventional military operations and makes them one of the most significant security challenges facing the international community today.

3. What is Critical Undersea Infrastructure?

Critical Undersea Infrastructure (CUI) refers to the network of physical assets, installations, and systems located on or beneath the seabed that are indispensable to the functioning of modern societies, economies, and national security. These assets facilitate global communications, energy production and transportation, electricity transmission, military operations, scientific research, and disaster preparedness. Their disruption—whether through natural disasters, accidents, cyber incidents, or deliberate acts of sabotage—can produce widespread economic losses, interrupt essential public services, undermine military readiness, and threaten international peace and security.

Unlike conventional terrestrial infrastructure, much of this infrastructure lies hundreds or even thousands of metres below the ocean surface. Its remote location makes continuous monitoring difficult, repairs technically complex, and attribution of malicious acts particularly challenging. As globalization has increased dependence on digital connectivity and cross-border energy networks, the protection of critical undersea infrastructure has become an increasingly important element of national resilience and international security policy.

Critical undersea infrastructure can be broadly classified into six major categories: communication infrastructure, energy infrastructure, pipelines, power transmission infrastructure, military infrastructure, and scientific infrastructure.

Undersea Communication Infrastructure

Submarine communication infrastructure forms the backbone of the global digital economy. Although satellites are often associated with international communications, the overwhelming majority of intercontinental data transmission occurs through submarine fibre-optic cables laid across the ocean floor. These systems enable internet access, cloud computing, financial transactions, diplomatic communications, and military command and control.

Internet Cables

Submarine internet cables are long-distance telecommunications cables installed along the seabed to connect countries and continents. These cables carry nearly all international internet traffic, enabling billions of daily digital interactions. They transmit emails, voice communications, video conferencing, financial transactions, and data exchanged between cloud servers and users worldwide.

Modern submarine cable networks are owned and operated by consortia consisting of telecommunications providers, technology companies, and governments. Because multiple countries often depend on a single cable system, damage to one cable can affect connectivity across entire regions.

Fibre-Optic Systems

The core component of submarine communication infrastructure is the fibre-optic system. These systems consist of extremely thin strands of highly purified glass or plastic through which data is transmitted as pulses of laser light.

Compared to satellite communications, fibre-optic systems offer significantly greater bandwidth, lower latency, higher reliability, and lower operational costs. Modern submarine fibre-optic cables can transmit several hundred terabits of data per second, supporting everything from internet browsing to international banking and secure government communications.

To withstand the harsh marine environment, submarine fibre-optic cables incorporate multiple protective layers, including waterproof insulation, steel armouring in shallow waters, copper conductors for power transmission, and durable outer sheathing.

Repeaters

As optical signals travel through fibre-optic cables, they gradually weaken over long distances. Repeaters, also known as optical amplifiers, are specialised devices installed along submarine cables—typically every 50 to 100 kilometres—to restore signal strength without converting the optical signal into electrical form.

Repeaters are powered through electrical currents carried within the submarine cable itself and operate continuously throughout the cable's lifespan, which often exceeds twenty-five years. The failure of a repeater can significantly reduce transmission capacity or interrupt communications until repairs are completed.

Landing Stations

Landing stations are secure coastal facilities where submarine cables emerge from the seabed and connect with terrestrial telecommunications networks. These facilities house equipment

responsible for powering submarine cables, managing data transmission, monitoring network performance, and directing traffic to domestic communication networks.

Landing stations are among the most strategically important components of submarine communication systems because they serve as gateways between international and national networks. Their concentration in relatively few coastal locations also makes them attractive targets for cyberattacks, espionage, and physical sabotage.

Energy Infrastructure

Energy infrastructure located offshore plays an essential role in global energy production and distribution. Offshore facilities extract hydrocarbons, process natural gas, and increasingly generate renewable energy through offshore wind farms.

Offshore Oil Platforms

Offshore oil platforms are large marine structures designed to drill, extract, and sometimes process crude oil from beneath the seabed. Depending on water depth and geological conditions, these platforms may be fixed to the seabed or operate as floating production systems.

Oil extracted from offshore fields is transported either through subsea pipelines or by tanker vessels to refineries and export terminals. Many countries rely heavily on offshore oil production, making these installations strategically significant economic assets.

Offshore Gas Platforms

Offshore gas platforms perform similar functions but extract natural gas from offshore reservoirs. Following extraction, natural gas may be transported through submarine pipelines directly to consumers or processed into liquefied natural gas (LNG) for international export.

Because natural gas plays an important role in electricity generation, industrial production, and domestic heating, offshore gas infrastructure constitutes a critical component of many national energy systems.

LNG Terminals

Liquefied Natural Gas (LNG) terminals receive, process, store, or export natural gas that has been cooled to approximately -162°C , reducing its volume by about 600 times and making long-distance maritime transport economically viable.

LNG terminals may be land-based or floating offshore facilities. They allow countries without pipeline connections to import natural gas from global markets, enhancing energy diversification and reducing dependence on individual suppliers.

Due to the hazardous nature of LNG and its strategic importance, these facilities require extensive physical, cyber, and environmental protection measures.

Offshore Wind Farms

Offshore wind farms consist of arrays of wind turbines installed in coastal or offshore waters to generate renewable electricity. Electricity generated by the turbines is transmitted to shore through submarine power cables before entering national electricity grids.

Compared with onshore wind installations, offshore wind farms generally experience stronger and more consistent wind conditions, allowing higher electricity generation. As many countries pursue decarbonisation and energy transition policies, offshore wind infrastructure has become an increasingly important component of national energy security.

Pipelines

Submarine pipelines transport hydrocarbons between offshore production facilities, coastal processing plants, and neighbouring countries. Their continuous operation is essential for maintaining stable energy supplies.

Gas Pipelines

Subsea gas pipelines transport natural gas across long distances beneath the ocean floor. These pipelines often connect offshore gas fields directly to mainland consumers or facilitate cross-border energy trade between neighbouring states.

Because many national electricity systems depend heavily on natural gas, disruptions to major gas pipelines can produce immediate economic and political consequences.

Oil Pipelines

Subsea oil pipelines transport crude oil from offshore extraction platforms to refineries, storage facilities, or export terminals. Compared with tanker transport, pipelines provide a continuous and cost-effective means of moving large quantities of petroleum.

Damage to oil pipelines can interrupt production, generate substantial economic losses, and cause severe marine pollution through oil spills.

Power Transmission Infrastructure

Growing international electricity interconnection has led to the rapid expansion of submarine power cable networks linking neighbouring countries and offshore renewable energy installations.

Cross-Border Electricity Cables

Cross-border electricity cables are submarine power transmission lines connecting the electricity grids of two or more countries. These interconnectors enable countries to exchange electricity, balance fluctuations in supply and demand, improve grid stability, and integrate renewable energy sources.

They contribute significantly to regional energy resilience by allowing electricity to flow where it is most needed during shortages or emergencies.

High-Voltage Direct Current (HVDC) Cables

High-Voltage Direct Current (HVDC) cables are specialised submarine electricity cables designed for efficient long-distance transmission of large quantities of electrical power. Compared with conventional alternating current (AC) systems, HVDC technology experiences lower transmission losses over long distances and is therefore preferred for submarine applications.

HVDC cables are widely used to connect offshore wind farms, remote islands, and national electricity grids separated by seas.

Military Infrastructure

Military undersea infrastructure provides surveillance, communication, and early warning capabilities essential for naval operations and national defence.

Naval Sensor Networks

Naval sensor networks consist of interconnected underwater sensors capable of detecting submarines, surface vessels, underwater vehicles, and other maritime activities. These systems utilise acoustic, magnetic, seismic, and pressure sensors to monitor strategic maritime areas and provide early warning of potential threats.

Such networks significantly enhance Maritime Domain Awareness (MDA) by improving situational awareness beneath the ocean surface.

SOSUS (Sound Surveillance System)

The Sound Surveillance System (SOSUS) is one of the earliest large-scale underwater surveillance networks, developed by the United States during the Cold War. Using arrays of seabed-mounted hydrophones, SOSUS was designed to detect and track Soviet submarines across the Atlantic and Pacific Oceans through long-range acoustic monitoring.

Although portions of the original system have been modernised or declassified, SOSUS remains an important historical example of strategic undersea surveillance and influenced the development of contemporary underwater monitoring technologies.

Underwater Surveillance Arrays

Modern underwater surveillance arrays are more advanced successors to early systems such as SOSUS. They combine seabed sensors, autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), underwater drones, sonar systems, and artificial intelligence to monitor underwater activity in real time.

These systems are increasingly employed to detect submarine movements, protect critical infrastructure, monitor maritime boundaries, and identify potential sabotage or unauthorized seabed operations.

Military Communication Cables

Military communication cables are dedicated submarine telecommunications systems used exclusively by armed forces and national security agencies. They provide secure, encrypted, and highly reliable communications between naval bases, command centres, strategic headquarters, and deployed military units.

Unlike commercial communication cables, military networks prioritise operational security, redundancy, and resilience against cyber and physical attacks, making them essential components of national defence infrastructure.

Scientific Infrastructure

Scientific undersea infrastructure supports oceanographic research, environmental monitoring, disaster risk reduction, and climate science. Although civilian in nature, these systems provide information critical for public safety and informed policymaking.

Ocean Observatories

Ocean observatories are permanent or semi-permanent underwater research facilities equipped with sensors that continuously monitor physical, chemical, geological, and biological conditions within the marine environment.

These observatories provide valuable long-term data on ocean circulation, marine ecosystems, seismic activity, underwater volcanism, and biodiversity, supporting both scientific research and evidence-based policymaking.

Climate Sensors

Climate sensors deployed beneath the oceans measure variables including water temperature, salinity, ocean currents, dissolved oxygen, carbon dioxide concentrations, and sea-level changes.

Because the oceans absorb approximately 90% of the excess heat generated by global warming, these sensors play a crucial role in understanding climate change, improving weather forecasting, monitoring marine ecosystems, and informing international climate policy.

Tsunami Warning Systems

Tsunami warning systems are integrated networks of seabed pressure sensors, ocean buoys, seismic monitoring stations, satellites, and coastal communication centres designed to detect undersea earthquakes and rapidly identify tsunami formation.

When abnormal pressure changes indicating a tsunami are detected, data are transmitted in real time to regional warning centres, which analyse the information and issue alerts to national authorities and vulnerable coastal communities. These systems have become an essential component of global disaster risk reduction and have significantly improved the ability of states to respond to tsunami threats following major seismic events.

Part II – Undersea Communication Cables

Introduction

Submarine communication cables are the physical foundation of the modern digital world. Although satellites are often perceived as the primary means of international communication, nearly all global digital connectivity depends on a vast network of fibre-optic cables laid across the ocean floor. These cables facilitate internet access, international telephone communications, cloud computing, financial transactions, military communications, diplomatic exchanges, scientific collaboration, and global commerce.

Today, submarine cables are recognised not merely as telecommunications infrastructure but as **critical strategic assets**. Their disruption can affect multiple countries simultaneously, interrupt economic activity, compromise military communications, and undermine national security. As geopolitical competition intensifies, these cables have become increasingly important within discussions on hybrid warfare, infrastructure resilience, and international maritime security.

Global Submarine Cable Network

The modern submarine cable network is one of the largest engineering systems ever constructed. According to current industry estimates, **more than 600 active and planned submarine cable systems** span the world's oceans, collectively extending **over 1.5 million kilometres**—enough to circle the Earth nearly forty times. These cables connect every inhabited continent except Antarctica and form the backbone of the global internet.

Contrary to popular belief, satellites account for only a small proportion of international communications. **Approximately 95–99% of all intercontinental internet traffic** is transmitted through submarine fibre-optic cables due to their significantly greater bandwidth, lower latency, and higher reliability. They support virtually every aspect of the digital economy, including web browsing, video conferencing, cloud computing, e-commerce, and government communications.

Submarine cables are equally indispensable to the global financial system. International payment networks, foreign exchange markets, securities trading platforms, and banking systems rely on these cables to transmit enormous volumes of data with minimal delay. It is estimated that **financial transactions worth several trillion US dollars every day** pass through submarine cable networks, meaning that even temporary disruptions can have immediate economic consequences.

Cloud computing has further increased dependence on submarine infrastructure. Major cloud service providers—including Google Cloud, Microsoft Azure, Amazon Web Services (AWS), and Meta's global infrastructure—operate interconnected data centres across continents. Submarine cables enable rapid data transfer between these facilities, allowing users

worldwide to access cloud storage, artificial intelligence services, streaming platforms, and enterprise applications in real time.

As digital transformation accelerates and global economies become increasingly interconnected, submarine cable networks have evolved from commercial telecommunications infrastructure into essential components of national resilience and international security.

How Submarine Cables Work

Although submarine communication systems span thousands of kilometres beneath the ocean, their underlying principle is relatively straightforward. Information is converted into pulses of light and transmitted through extremely thin strands of glass known as optical fibres. These light signals travel at very high speeds with minimal loss, allowing enormous quantities of information to move almost instantaneously between continents.

Fibre Optics

At the heart of every submarine cable lies a bundle of optical fibres, each approximately the thickness of a human hair. Rather than carrying electricity, these fibres transmit laser light. Information—whether an email, financial transaction, phone call, or video stream—is encoded into light pulses that travel through the fibre by repeated internal reflection.

Modern fibre-optic technology provides exceptionally high bandwidth and low latency, making it significantly faster and more efficient than satellite communications for long-distance data transmission. A single modern cable can carry hundreds of terabits of data per second, serving millions of users simultaneously.

To survive the harsh marine environment, the optical fibres are protected by multiple layers of waterproof insulation, steel armouring, copper conductors, and polyethylene sheathing. In shallow coastal waters, where fishing activity and ship anchors pose greater risks, cables receive additional protective armour.

Repeaters

As light travels through optical fibres, the signal gradually weakens due to attenuation. To maintain signal quality across thousands of kilometres, **optical repeaters** are installed along the cable, typically every **50–100 kilometres**.

Unlike traditional electronic repeaters, modern optical repeaters amplify the light signal directly without converting it into electrical form, thereby reducing transmission delays and improving reliability. These repeaters are powered through electrical current supplied continuously from cable landing stations located at either end of the cable.

Failure of a single repeater can reduce communication capacity or interrupt an entire cable segment until specialized repair operations are completed.

Cable Landing Stations

Cable landing stations are highly secure coastal facilities where submarine cables transition from the seabed to terrestrial communication networks. These stations perform several essential functions, including supplying electrical power to the submarine cable, monitoring network performance, managing traffic flows, and connecting international communications to domestic telecommunications infrastructure.

Landing stations often house sophisticated network equipment and redundant systems designed to ensure continuous operation. Because many cables converge at a limited number of landing stations, these facilities represent strategic points of vulnerability. Physical attacks, cyber intrusions, or natural disasters affecting landing stations can disrupt communications even if the submarine cable itself remains undamaged.

Branching Units

Many submarine cable systems do not simply connect two countries; instead, they serve multiple destinations through **branching units** installed on the seabed. These specialised devices divide a single main cable into several branches, allowing one cable system to connect multiple countries or regions simultaneously.

Branching units improve the efficiency and cost-effectiveness of global cable networks by enabling a single transoceanic system to provide connectivity across numerous coastal states. However, because multiple communication routes may depend upon a single branching unit, its failure can affect several countries simultaneously.

Cable Repair Ships

Despite their durability, submarine cables are occasionally damaged by fishing activity, ship anchors, earthquakes, underwater landslides, equipment failures, or deliberate interference. When a fault occurs, specialised **cable repair ships** are deployed to locate and repair the damaged section.

Using remotely operated vehicles (ROVs), grappling equipment, sonar systems, and precision navigation technologies, repair crews recover the damaged cable from the seabed, splice new sections into the cable, thoroughly test the repaired connection, and carefully relay it on the ocean floor.

Repairs in shallow coastal waters may require only several days, whereas faults occurring in deep oceans or adverse weather conditions can require several weeks. Because only a limited number of specialised repair vessels operate worldwide, simultaneous cable failures may significantly delay restoration efforts.

Ownership of Submarine Cables

Unlike many forms of critical infrastructure, submarine communication cables are not owned exclusively by governments. Instead, ownership is distributed among private technology companies, telecommunications providers, investment consortia, and, in some cases, states.

Historically, submarine cables were financed and operated primarily by national telecommunications companies. However, the rapid growth of cloud computing, social media, and digital services has fundamentally transformed ownership patterns.

Technology Companies

Large technology companies have become some of the world's largest investors in submarine cable infrastructure.

- **Google** owns or co-owns numerous transoceanic cable systems connecting North America, Europe, Asia, Africa, and South America to support Google Cloud, Search, YouTube, and AI services.
- **Meta** has invested heavily in global submarine cable projects to support Facebook, Instagram, WhatsApp, and future digital infrastructure initiatives.
- **Microsoft** participates in submarine cable investments primarily to strengthen the global connectivity of Microsoft Azure and enterprise cloud services.
- **Amazon**, through Amazon Web Services (AWS), invests in submarine infrastructure to improve connectivity between global data centres and cloud customers.

Rather than relying solely on third-party telecommunications providers, these companies increasingly invest directly in cable infrastructure to ensure higher capacity, lower latency, improved reliability, and greater operational control.

Telecommunications Companies

National and international telecommunications operators remain among the principal owners of submarine cable systems. Companies such as Orange, AT&T, Vodafone, NTT, China Telecom, Singtel, Tata Communications, and many others own or operate significant portions of the global network.

These companies traditionally provide international telecommunications services and continue to play a central role in financing, operating, and maintaining cable systems.

Consortium Ownership

Most submarine cables are not owned by a single organization. Instead, they are financed through **international consortia**, in which several telecommunications companies, technology firms, or governments jointly share construction costs, maintenance responsibilities, and transmission capacity.

Consortium ownership distributes financial risk while allowing participating members to secure dedicated communication capacity. It has become the predominant model for large intercontinental cable projects.

Government Ownership

Governments generally own fewer commercial submarine cables than private companies. However, states frequently own military communication cables, strategic government networks, or portions of infrastructure deemed essential to national security.

Governments also regulate cable landings, issue operating licences, oversee cybersecurity standards, and increasingly review foreign investment in submarine infrastructure due to national security concerns.

The Trend Towards Private Ownership

Over the past decade, ownership has shifted significantly from traditional telecommunications operators toward global technology companies. Today, hyperscale cloud providers finance an increasing proportion of new submarine cable projects.

This trend reflects the growing importance of cloud computing, artificial intelligence, and digital services but also raises important policy questions concerning market concentration, data sovereignty, infrastructure resilience, cybersecurity, and the governance of privately owned assets that perform essential public functions.

Global Cable Chokepoints

Submarine cables do not cross the oceans uniformly. Instead, they are concentrated along specific maritime corridors and narrow waterways known as **chokepoints**, where numerous cable systems converge. These locations are strategically significant because disruption at a single chokepoint can affect multiple cable systems simultaneously, amplifying the impact on global communications.

English Channel

The English Channel is one of the busiest maritime corridors in the world and serves as a critical connection between the United Kingdom and mainland Europe. Numerous submarine cables carrying internet, financial, and telecommunications traffic pass through this narrow body of water. Its dense concentration of infrastructure, combined with intense shipping activity, makes it vulnerable to accidental damage and potential sabotage.

Mediterranean Sea

The Mediterranean acts as a bridge between Europe, North Africa, and the Middle East. Many cable systems linking Europe with Asia and Africa traverse this enclosed sea before

continuing through the Suez Canal and Red Sea. Any disruption here could affect communications across three continents.

Red Sea

The Red Sea is among the world's most critical submarine cable corridors. A large proportion of Europe–Asia internet traffic passes through this narrow maritime route before entering the Mediterranean via the Suez Canal. Recent regional instability has highlighted the vulnerability of cables in this area to both accidental damage and conflict-related disruptions.

Bab-el-Mandeb Strait

Connecting the Red Sea with the Gulf of Aden, the Bab-el-Mandeb Strait is a narrow maritime passage through which numerous submarine cables converge. Due to its geographic constraints and proximity to regional conflicts, it represents one of the world's most strategically sensitive communication chokepoints.

Strait of Malacca

The Strait of Malacca links the Indian Ocean with the Pacific Ocean and supports extensive shipping and submarine cable traffic connecting Europe with East Asia. Its heavy maritime traffic increases the risk of anchor damage, while its strategic importance makes it a focal point in regional security discussions.

South China Sea

The South China Sea hosts numerous submarine cable systems connecting East and Southeast Asia. Ongoing territorial disputes, increasing naval activity, and strategic competition among regional powers have heightened concerns regarding the security of cable infrastructure in this region.

Taiwan Strait

The Taiwan Strait serves as a vital communication corridor linking Taiwan with neighbouring economies and global internet networks. Repeated cable disruptions and heightened geopolitical tensions have made this area an important case study in infrastructure resilience and hybrid threats.

North Atlantic

The North Atlantic contains many of the principal cable routes connecting North America and Europe. These cables support a significant proportion of transatlantic financial transactions, military communications, and cloud services. Their strategic importance has led to increased naval surveillance and infrastructure protection efforts.

Arctic Routes

Climate change and diminishing sea ice are gradually opening new Arctic shipping lanes and creating opportunities for future submarine cable routes between Europe, North America, and Asia. While these routes may reduce transmission distances, they also introduce new challenges relating to harsh environmental conditions, limited repair capabilities, geopolitical competition, and governance in the Arctic.

Vulnerabilities of Submarine Cables

Despite being engineered for durability, submarine cables remain exposed to a wide range of natural, accidental, and deliberate threats.

- **Fishing activity:** Trawling gear and fishing nets can snag or sever cables in shallow waters, making fishing one of the most common causes of cable damage.
- **Ship anchors:** Anchors dragged across the seabed, particularly during storms or emergencies, can break or damage cables, especially near coastlines and busy ports.
- **Earthquakes:** Seismic activity may fracture the seabed, displace cable routes, or trigger underwater landslides, resulting in multiple simultaneous cable failures.
- **Volcanic activity:** Underwater volcanic eruptions and associated geological changes can destroy cables, alter seabed topography, and complicate repair operations.
- **Submarine landslides:** Large-scale movements of sediment on the ocean floor can bury, stretch, or break cables over extensive distances, sometimes causing regional communication outages.
- **Espionage:** Intelligence agencies may attempt to intercept communications, map cable routes, or monitor traffic through sophisticated underwater operations. Even when data is encrypted, metadata and physical access to infrastructure may hold strategic value.
- **Cyber attacks:** While fibre-optic cables themselves are difficult to hack directly, associated infrastructure—including landing stations, network management systems, and cloud services—may be targeted to disrupt communications or manipulate traffic.
- **Sabotage:** Deliberate physical damage to cables using divers, autonomous underwater vehicles, submersibles, explosives, or disguised commercial vessels has become a growing security concern, particularly during periods of geopolitical tension.
- **Military attacks:** During armed conflict, submarine communication cables may be targeted to disrupt command-and-control systems, intelligence sharing, logistics, and civilian communications. Such attacks could significantly affect both military operations and civilian populations.
- **Insider threats:** Individuals with authorised access to cable infrastructure, maintenance systems, or operational information may intentionally or unintentionally compromise security, highlighting the importance of personnel vetting and internal controls.
- **Supply-chain attacks:** The increasingly complex global supply chain for cable manufacturing, software, network equipment, and maintenance introduces risks of

tampered hardware, compromised software, counterfeit components, or hidden vulnerabilities that could be exploited before infrastructure even becomes operational.

Collectively, these vulnerabilities illustrate why submarine communication cables are no longer viewed solely as commercial telecommunications assets. They have become strategic infrastructure whose protection requires coordinated action by governments, private industry, international organizations, and regional security frameworks.

Part III – Hybrid Threats in Detail

Introduction

Hybrid threats encompass a broad spectrum of coordinated activities that combine military and non-military means to undermine an adversary without necessarily crossing the threshold of conventional armed conflict. In the maritime domain, these threats target critical undersea infrastructure to disrupt communications, energy supplies, military readiness, and economic stability while maintaining ambiguity regarding responsibility.

Unlike traditional warfare, hybrid operations often involve multiple actors—including state agencies, proxy groups, private companies, cybercriminals, and civilian vessels—acting simultaneously across physical, cyber, informational, economic, and intelligence domains. This complexity makes hybrid attacks particularly difficult to detect, attribute, and deter.

Critical undersea infrastructure presents an attractive target because it is geographically dispersed, technically complex, privately owned in many cases, and difficult to monitor continuously. The following sections examine the principal forms of hybrid threats relevant to submarine communication cables and other undersea infrastructure.

Physical Sabotage

Physical sabotage refers to the deliberate destruction, damage, or interference with critical infrastructure through direct physical means. Unlike cyber operations, physical sabotage aims to impair infrastructure by causing tangible damage to cables, pipelines, sensors, or offshore installations.

Because undersea infrastructure is located in remote and difficult-to-monitor environments, identifying deliberate sabotage can be challenging. Physical damage may initially resemble accidental causes such as anchor strikes or natural disasters, allowing perpetrators to exploit uncertainty and delay attribution.

Various platforms and actors may be employed to conduct physical sabotage.

Divers

Combat divers are capable of operating in relatively shallow waters to inspect, tamper with, or place explosive charges on underwater infrastructure. While their operational depth is limited compared to modern underwater vehicles, divers remain effective against coastal infrastructure, harbour facilities, and cable landing areas.

Mini-Submarines

Mini-submarines, also known as midget submarines, are small crewed underwater vessels capable of transporting personnel and equipment covertly. Their compact size allows them to operate in confined waters while conducting surveillance, deploying divers, or placing devices on undersea infrastructure.

Some states maintain specialised mini-submarines for intelligence and special operations, making them potential tools for covert infrastructure interference.

Remotely Operated Vehicles (ROVs)

ROVs are tethered underwater vehicles controlled from surface vessels. Although primarily designed for commercial inspection, offshore engineering, and scientific research, they possess the precision necessary to inspect, manipulate, repair, or potentially damage undersea infrastructure.

Because ROVs are widely used in legitimate offshore industries, distinguishing lawful commercial activity from malicious operations may be difficult.

Unmanned Underwater Vehicles (UUVs)

Unlike ROVs, Unmanned Underwater Vehicles (UUVs) operate autonomously or semi-autonomously without a physical tether to a support vessel. Advances in artificial intelligence, navigation systems, and battery technology have significantly expanded their operational range and capabilities.

UUVs may conduct seabed mapping, intelligence collection, infrastructure inspection, or surveillance. In military contexts, concerns have emerged regarding their potential use for covert sabotage, mine deployment, or interference with submarine cables and pipelines.

Naval Special Forces

Many states maintain specialised naval units trained to conduct underwater reconnaissance, demolition, sabotage, and covert maritime operations. These forces possess advanced diving equipment, underwater navigation systems, and explosive expertise, enabling them to operate against strategic maritime targets.

Because such operations are typically highly classified, establishing responsibility for suspected state involvement presents considerable political and legal challenges.

Merchant Vessels

Commercial cargo ships may unintentionally damage submarine cables through anchor dragging or other maritime activities. However, concerns have grown regarding the possibility that civilian merchant vessels could deliberately be used to interfere with undersea infrastructure while appearing to conduct routine commercial operations.

The dual-use nature of merchant shipping complicates investigations and attribution, particularly in heavily trafficked maritime corridors.

Fishing Vessels

Fishing vessels account for a significant proportion of accidental cable damage through bottom trawling and fishing gear. Nevertheless, they may also be employed as platforms for surveillance, intelligence gathering, or deliberate interference, particularly when operating under government direction.

Because fishing vessels routinely operate in areas containing submarine cables, distinguishing between accidental incidents and intentional actions may require extensive forensic investigation.

Cyber Attacks

Although submarine cables themselves are physical assets, the systems responsible for operating, managing, and routing global communications depend heavily on digital infrastructure. Consequently, cyberattacks may achieve widespread disruption without physically damaging the cables.

Landing Stations

Cable landing stations contain critical network equipment responsible for managing international communications. Cyber intrusions targeting these facilities may disrupt traffic management, disable monitoring systems, or compromise operational control.

Given their concentration of network infrastructure, landing stations represent high-value cyber targets.

Network Management Systems

Modern submarine cable systems are supervised through sophisticated network management platforms that continuously monitor signal quality, equipment status, power supply, and traffic routing.

Cyberattacks against these systems may manipulate operational data, delay fault detection, or interfere with maintenance activities, reducing network resilience.

Satellite Integration

Although most international communications travel through submarine cables, satellite systems remain essential for backup communications, navigation, timing, and operational monitoring.

Cyberattacks targeting integrated satellite services may reduce redundancy during cable disruptions, complicating restoration efforts and emergency response.

Routing Systems

Internet routing determines how digital information travels between networks worldwide. Malicious actors may exploit routing protocols through techniques such as route hijacking or route manipulation, redirecting traffic to unintended destinations, enabling surveillance, or causing service disruptions.

Such attacks can significantly affect internet availability without damaging physical infrastructure.

Internet Exchange Points (IXPs)

Internet Exchange Points are facilities where internet service providers exchange data traffic efficiently. Cyberattacks targeting IXPs may degrade connectivity across multiple networks simultaneously, amplifying the effects of disruptions elsewhere in the communication system.

Because many international communications pass through major IXPs, their protection forms an essential component of digital infrastructure resilience.

Electronic Warfare

Electronic warfare involves the military use of the electromagnetic spectrum to disrupt, deceive, or deny an adversary's communications, navigation, and sensing capabilities. While it does not directly damage physical infrastructure, it can significantly impair maritime operations.

GPS Jamming

Global Positioning System (GPS) jamming involves transmitting powerful radio signals that prevent legitimate GPS receivers from determining accurate positions.

In maritime environments, GPS jamming may affect commercial shipping, offshore maintenance operations, cable repair vessels, and military navigation, increasing operational risks.

AIS Spoofing

The Automatic Identification System (AIS) enables vessels to broadcast their identity, position, speed, and course to improve maritime safety.

AIS spoofing involves transmitting false vessel information or manipulating location data, causing ships to appear in incorrect locations or conceal their true movements. Such deception complicates maritime surveillance and may obscure suspicious activity near critical infrastructure.

Signal Interference

Signal interference refers to the intentional disruption of radio communications through electromagnetic transmissions that reduce communication quality or prevent reliable information exchange.

Critical maritime communications, offshore platforms, emergency response systems, and repair operations may all be affected by signal interference.

Communications Denial

Communications denial encompasses broader efforts to prevent adversaries from transmitting or receiving information. This may involve jamming radio frequencies, disrupting satellite communications, interfering with data links, or targeting communication infrastructure through cyber or electronic means.

In combination with physical sabotage, communications denial may significantly delay detection and response following attacks on critical infrastructure.

Intelligence Gathering

Undersea infrastructure represents both a communications medium and a valuable source of intelligence. Consequently, many states devote significant resources to monitoring submarine cable systems and maritime activity.

Wire Tapping

Historically, intelligence agencies have developed specialised technologies capable of accessing communications travelling through submarine cables. Modern fibre-optic systems employ strong encryption, making direct interception considerably more difficult than in earlier decades. Nevertheless, concerns remain regarding unauthorized access to communications infrastructure.

Cable Interception

Cable interception involves obtaining physical or electronic access to submarine communication systems for intelligence purposes. While modern technology has improved cable security, advances in underwater vehicles and specialised engineering capabilities continue to raise concerns regarding potential interception operations.

Signal Interception

Signal interception refers to the collection of electromagnetic emissions associated with communication systems, radar installations, satellite links, or other electronic transmissions. Such intelligence supports military planning, situational awareness, and strategic decision-making.

Ocean Surveillance

Ocean surveillance encompasses the continuous monitoring of maritime activity using satellites, sonar systems, underwater sensors, autonomous vehicles, aircraft, and surface vessels.

Effective ocean surveillance enables states to detect suspicious activities near critical infrastructure, monitor submarine movements, and enhance maritime domain awareness.

Acoustic Monitoring

Sound travels efficiently through seawater, making acoustic monitoring one of the principal methods of underwater surveillance.

Hydrophones and seabed sensor networks continuously detect underwater sounds produced by ships, submarines, underwater vehicles, and other activities. Acoustic monitoring supports anti-submarine warfare, infrastructure protection, and environmental observation.

Information Warfare

Physical attacks on infrastructure are frequently accompanied by campaigns designed to influence public perception, delay attribution, or manipulate political responses.

Disinformation Following Cable Cuts

Following cable disruptions, false or misleading information may be intentionally disseminated regarding the cause of the incident, the extent of damage, or the parties responsible.

Such disinformation may create confusion, reduce public trust, and complicate emergency response efforts.

False Attribution

False attribution involves deliberately assigning responsibility for an incident to another state or actor without sufficient evidence. In hybrid conflicts, competing narratives often emerge immediately after infrastructure incidents, making impartial investigations essential.

Premature or inaccurate attribution may heighten diplomatic tensions and increase the risk of escalation.

Media Manipulation

Media manipulation involves influencing traditional or digital media to shape public understanding of events. Selective reporting, coordinated online campaigns, fabricated evidence, and automated social media activity may all be employed to reinforce strategic narratives.

Psychological Warfare

Psychological warfare seeks to influence the attitudes, morale, and decision-making of governments, military personnel, and civilian populations. Following attacks on critical infrastructure, exaggerated claims regarding vulnerability or state incapacity may be used to undermine public confidence and increase political pressure.

Economic Coercion

Economic instruments increasingly form part of hybrid strategies by imposing financial costs without resorting to military force.

Blocking Repairs

Following damage to submarine infrastructure, access restrictions, denial of repair permits, interference with repair vessels, or prolonged diplomatic disputes may delay restoration efforts. Such delays can extend economic disruption well beyond the initial incident.

Insurance Pressure

Marine insurance plays a crucial role in financing commercial shipping and infrastructure operations. Increased insurance premiums or withdrawal of coverage in high-risk regions may discourage investment, increase operating costs, and reduce commercial activity.

Insurance markets can therefore indirectly influence strategic maritime behaviour.

Sanctions

Economic sanctions targeting technology exports, telecommunications equipment, offshore engineering services, or financial institutions may affect the construction, maintenance, and resilience of critical undersea infrastructure.

Sanctions have become an increasingly common instrument of statecraft within broader hybrid competition.

Supply-Chain Disruption

The construction and maintenance of submarine cables depend upon globally distributed supply chains involving specialised manufacturers, optical components, semiconductors, cable-laying vessels, repair equipment, and engineering expertise.

Disruptions affecting any component of this supply chain—whether through export controls, geopolitical tensions, cyber incidents, or transportation interruptions—may delay infrastructure development, increase costs, and reduce operational resilience.

Investment Leverage

Control over investment in critical infrastructure can itself become a source of strategic influence. States or state-backed enterprises may finance submarine cable projects, acquire ownership stakes, or provide infrastructure loans that increase their economic and political leverage over recipient countries.

Such investments may offer substantial developmental benefits while simultaneously raising concerns regarding strategic dependence, data security, infrastructure governance, and long-term geopolitical influence. For this reason, many governments have strengthened foreign investment screening mechanisms for critical infrastructure projects.

Part IV – Geopolitical Dimension

Introduction

Critical undersea and maritime infrastructure has become a central element of contemporary geopolitics. Unlike traditional maritime security, which primarily focused on naval power and sea lanes, modern geopolitical competition increasingly extends beneath the ocean surface, where submarine communication cables, energy pipelines, offshore energy installations, military surveillance systems, and power interconnectors underpin the global economy and national security.

The growing interdependence of digital communications, global supply chains, and energy networks means that disruptions to undersea infrastructure can have far-reaching consequences beyond the immediate region in which they occur. As competition among major powers intensifies, the protection, ownership, surveillance, and governance of undersea infrastructure have emerged as strategic priorities.

Today, critical undersea infrastructure is no longer viewed solely as civilian infrastructure; it is increasingly regarded as a strategic asset capable of influencing diplomacy, military operations, economic resilience, and technological competition.

Strategic Competition

United States–China

The strategic rivalry between the United States and China increasingly extends into the digital and maritime domains. Both countries recognize submarine communication cables as essential infrastructure supporting military communications, economic activity, cloud computing, artificial intelligence, and technological innovation.

The United States has expanded investment in secure submarine cable systems among allied nations while increasing scrutiny of foreign involvement in critical telecommunications infrastructure. National security concerns have led to greater oversight of cable landing licences, foreign investment, and supply-chain security.

China, through initiatives such as the **Digital Silk Road**, has invested heavily in global digital infrastructure, including submarine cable projects, telecommunications equipment, and data connectivity across Asia, Africa, Latin America, and Europe. Chinese technology companies have sought participation in international cable consortia, reflecting China's broader objective of strengthening digital connectivity and expanding its technological influence.

The resulting competition is not merely commercial but strategic, encompassing cybersecurity, supply-chain resilience, technological standards, data governance, and control over future digital infrastructure.

Russia–NATO

Relations between Russia and NATO have increasingly highlighted the strategic importance of critical undersea infrastructure. NATO member states rely extensively on submarine communication cables and offshore energy networks linking Europe and North America, while Russia possesses significant naval capabilities and experience operating in the Arctic, North Atlantic, Baltic Sea, and Black Sea.

Following the 2022 Nord Stream pipeline explosions and subsequent incidents affecting submarine cables and pipelines in European waters, concerns regarding infrastructure protection have intensified. NATO has increased surveillance, intelligence sharing, and naval patrols around critical undersea assets, while several European states have strengthened infrastructure monitoring capabilities.

Although attribution for individual incidents often remains contested, the broader geopolitical environment has reinforced the perception that undersea infrastructure constitutes a potential target during periods of strategic competition.

Indo-Pacific

The Indo-Pacific has emerged as one of the world's most strategically significant regions due to its concentration of maritime trade routes, submarine communication cables, and geopolitical rivalries.

Countries including India, Japan, Australia, the United States, China, and ASEAN member states depend heavily upon secure maritime infrastructure to sustain economic growth, energy imports, and digital connectivity. Rapid expansion of submarine cable systems, offshore energy development, and naval modernization has increased both the strategic value and vulnerability of regional infrastructure.

The region has consequently become an important focus for maritime security cooperation, infrastructure resilience initiatives, and investments in secure digital connectivity.

Arctic

Climate change is transforming the Arctic into an increasingly important geopolitical region. Retreating sea ice is opening new maritime routes and creating opportunities for future submarine communication cables connecting Europe, North America, and Asia through shorter northern passages.

The Arctic also contains substantial hydrocarbon resources, critical minerals, and strategic military positions. Consequently, Arctic coastal states have expanded scientific research, infrastructure development, military presence, and surveillance capabilities.

However, the region presents significant operational challenges due to extreme weather, limited repair capabilities, sparse infrastructure, and evolving legal questions concerning navigation and resource exploitation.

Mediterranean

The Mediterranean Sea functions as a major junction connecting Europe, North Africa, and the Middle East. Numerous submarine communication cables carrying traffic between Europe, Asia, and Africa traverse the Mediterranean before reaching the Red Sea through the Suez Canal.

In addition to digital infrastructure, the Mediterranean contains extensive offshore gas developments, electricity interconnectors, and strategic shipping routes. Regional conflicts, migration pressures, maritime boundary disputes, and competing energy interests contribute to its continuing geopolitical significance.

Red Sea

The Red Sea has become one of the world's most strategically sensitive maritime regions. It serves as the primary maritime corridor connecting Europe and Asia through the Suez Canal and contains one of the highest concentrations of submarine communication cables globally.

Recent regional conflicts, attacks on commercial shipping, and disruptions affecting submarine cables have highlighted the vulnerability of this corridor. Because numerous international communication systems converge within a relatively narrow geographic area, even localized incidents may produce widespread international consequences.

Indian Ocean

The Indian Ocean connects major energy producers in the Middle East with rapidly growing economies across Asia. It accommodates extensive shipping traffic, submarine communication cables, offshore energy infrastructure, and critical maritime chokepoints.

Growing naval modernization, increased commercial investment, expanding digital infrastructure, and competition among major powers have elevated the strategic importance of the region. Several states have strengthened maritime domain awareness, naval cooperation, and infrastructure protection initiatives throughout the Indian Ocean.

South China Sea

The South China Sea is among the world's busiest maritime regions, supporting substantial commercial shipping, fisheries, offshore energy exploration, and numerous submarine communication cable systems.

Competing territorial claims, extensive land reclamation, military deployments, and increasing strategic rivalry have made the region a focal point of international security discussions. Because many regional cable systems traverse disputed waters, infrastructure protection has become closely linked with broader questions of maritime governance and regional stability.

Baltic Sea

The Baltic Sea contains dense concentrations of submarine communication cables, gas pipelines, electricity interconnectors, and offshore energy infrastructure connecting Northern and Central Europe.

Following several infrastructure incidents since 2022—including damage to pipelines and communication cables—the Baltic has emerged as an important case study regarding infrastructure resilience, maritime surveillance, and attribution challenges. The region's relatively confined geography and extensive infrastructure make it particularly sensitive to both accidental and deliberate disruptions.

Black Sea

The Black Sea occupies a strategic position linking Eastern Europe, the Caucasus, and the Mediterranean. It supports commercial shipping, offshore energy activities, submarine communications, and naval operations by several littoral states.

Recent regional conflicts have increased military activity throughout the Black Sea and highlighted the vulnerability of maritime infrastructure to hybrid threats, including cyber operations, electronic warfare, and physical sabotage.

Maritime Chokepoints

Maritime chokepoints are narrow waterways through which significant proportions of global shipping, energy supplies, and submarine communication cables pass. Because traffic is concentrated within limited geographic areas, disruptions at these locations can have disproportionate international consequences.

Suez Canal

The Suez Canal connects the Mediterranean Sea with the Red Sea, providing the shortest maritime route between Europe and Asia. In addition to carrying approximately 10–12% of global maritime trade, the canal corridor accommodates numerous submarine communication cables linking Europe with Asia and East Africa.

Disruptions to the canal or surrounding infrastructure may significantly affect international shipping, digital communications, and global supply chains.

Panama Canal

The Panama Canal connects the Atlantic and Pacific Oceans, reducing transit times for commercial shipping between the eastern and western hemispheres. While it carries fewer submarine cables than the Suez corridor, it remains essential for global trade and maritime logistics.

Bab-el-Mandeb Strait

The Bab-el-Mandeb Strait links the Red Sea with the Gulf of Aden and serves as one of the world's most important communication and energy corridors. Multiple submarine cable systems converge within this narrow passage, making it a critical vulnerability in global internet connectivity.

Its proximity to regional conflicts further increases security concerns.

Strait of Hormuz

The Strait of Hormuz connects the Persian Gulf with the Gulf of Oman and the Arabian Sea. It represents one of the world's most important energy chokepoints, through which a substantial proportion of globally traded crude oil and liquefied natural gas passes.

While primarily associated with energy transportation, its strategic significance influences the broader security of regional offshore infrastructure.

Strait of Malacca

The Strait of Malacca connects the Indian Ocean with the South China Sea and the Pacific Ocean. It supports extensive commercial shipping and numerous submarine cable systems linking Europe with East Asia.

Its narrow geography, dense maritime traffic, and economic importance make it one of the world's most strategically significant maritime passages.

Taiwan Strait

The Taiwan Strait provides an essential communication corridor connecting Taiwan with international submarine cable networks. It also constitutes a strategically sensitive area due to persistent geopolitical tensions and repeated incidents affecting cable connectivity.

Its importance extends beyond regional communications, influencing broader Indo-Pacific digital resilience.

GIUK Gap

The Greenland–Iceland–United Kingdom (GIUK) Gap is a strategic maritime passage in the North Atlantic connecting the Arctic Ocean with the Atlantic. During the Cold War, it served as a critical anti-submarine defence line for NATO, enabling monitoring of Soviet naval movements.

Today, the GIUK Gap remains strategically important due to transatlantic submarine communication cables, naval operations, and growing Arctic maritime activity.

Maritime Jurisdiction Under International Law

The protection of undersea infrastructure depends heavily upon the legal status of the waters in which it is located. The **United Nations Convention on the Law of the Sea (UNCLOS)** establishes the principal legal framework governing maritime zones and the rights and responsibilities of states.

Territorial Waters

Territorial waters extend up to **12 nautical miles** from a coastal state's baseline. Within this zone, the coastal state exercises sovereignty comparable to that exercised over its land territory, including authority over the seabed, subsoil, water column, and airspace above.

States are therefore primarily responsible for protecting critical undersea infrastructure located within their territorial sea, although foreign vessels generally retain the right of innocent passage under UNCLOS.

Exclusive Economic Zone (EEZ)

The Exclusive Economic Zone extends up to **200 nautical miles** from the coastal baseline. Within the EEZ, the coastal state possesses **sovereign rights**—rather than full sovereignty—for the exploration, exploitation, conservation, and management of natural resources in the waters, seabed, and subsoil.

Coastal states may authorize offshore energy development, fisheries, and scientific research, while retaining jurisdiction over artificial islands, offshore installations, and environmental protection. However, other states continue to enjoy freedoms of navigation, overflight, and the laying and maintenance of submarine cables and pipelines, subject to the provisions of UNCLOS.

As many submarine communication cables and offshore energy installations traverse EEZs belonging to multiple states, cooperation and legal coordination are essential for infrastructure protection.

High Seas

The high seas comprise all maritime areas beyond national jurisdiction. Under UNCLOS, no state may claim sovereignty over the high seas, which remain open to all states for navigation, overflight, fishing, scientific research, and the laying of submarine cables and pipelines.

Because no single government exercises authority over these waters, protecting critical infrastructure on the high seas requires international cooperation, information sharing, and adherence to international law.

Continental Shelf

The continental shelf refers to the submerged natural prolongation of a coastal state's land territory. Under UNCLOS, coastal states possess sovereign rights over the exploration and exploitation of natural resources located on or beneath their continental shelf, including oil, gas, and mineral resources.

Importantly, these rights concern the seabed and subsoil rather than the water column above. Other states generally retain the freedom to lay submarine cables and pipelines across another state's continental shelf, although routing must respect the coastal state's rights regarding resource exploitation and environmental protection.

Deep Seabed (The Area)

The deep seabed—referred to in UNCLOS as "**the Area**"—comprises the seabed and ocean floor beyond the limits of national jurisdiction. It is designated as the **common heritage of humankind**, meaning that its mineral resources are managed collectively through the **International Seabed Authority (ISA)** for the benefit of humanity as a whole.

Although relatively little critical infrastructure currently exists in the deep seabed, future developments involving deep-sea mining, transoceanic cable systems, and scientific observatories are expected to increase its strategic importance, raising new legal, environmental, and security challenges for the international community.

Part V – Legal Framework

Introduction

The protection of critical undersea and maritime infrastructure is governed not by a single treaty but by a complex framework of international law, maritime conventions, cybersecurity principles, and technical standards. These legal instruments regulate the rights and obligations of states regarding submarine communication cables, pipelines, offshore installations, navigation, cybersecurity, and the peaceful use of the oceans.

Although many of these instruments were developed before the emergence of modern hybrid threats, they continue to provide the primary legal basis for protecting undersea infrastructure. However, recent incidents involving cable damage, cyberattacks, and covert maritime operations have revealed significant legal gaps concerning attribution, state responsibility, and responses to hostile activities below the threshold of armed conflict.

United Nations Convention on the Law of the Sea (UNCLOS)

Adopted in **1982** and entering into force in **1994**, the **United Nations Convention on the Law of the Sea (UNCLOS)** is widely regarded as the "Constitution of the Oceans." With more than 160 States Parties, it establishes the comprehensive legal framework governing maritime zones, navigation, marine resources, environmental protection, scientific research, and the laying and protection of submarine cables and pipelines.

While UNCLOS does not specifically address modern cyber threats or hybrid warfare, it remains the principal treaty governing activities affecting critical undersea infrastructure.

Freedom of Navigation

One of UNCLOS's fundamental principles is the **freedom of navigation**, particularly on the high seas and within Exclusive Economic Zones (EEZs). States enjoy the right to navigate, conduct overflight, lay submarine cables and pipelines, and undertake other internationally lawful uses of the sea, provided these activities respect international law and the rights of coastal states.

Freedom of navigation ensures that international communications, commerce, and maritime transportation can operate without arbitrary interference. It also enables the construction and maintenance of submarine communication networks connecting different regions of the world.

Protection of Submarine Cables

UNCLOS recognises the importance of submarine communication cables as essential infrastructure for international communications.

- **Article 58** preserves the freedom of all states to lay submarine cables and pipelines within the Exclusive Economic Zone, subject to the Convention.
- **Article 79** specifically provides that all states are entitled to lay submarine cables and pipelines on the continental shelf of other states. Coastal states may establish reasonable conditions concerning the routing of pipelines but generally may not prevent the laying or maintenance of submarine communication cables.
- **Articles 113–115** require States Parties to adopt domestic laws criminalising the intentional or negligent breaking or damaging of submarine cables and establish obligations relating to liability and repairs.

These provisions reflect the longstanding international recognition that submarine communication systems are global infrastructure serving the interests of all states.

Repair Rights

UNCLOS recognises that damaged submarine cables must be repaired promptly to preserve international communications.

States undertaking repair operations are generally entitled to access damaged cables, subject to the rights of coastal states and applicable national regulations. Coastal states are expected to facilitate lawful repair activities while balancing environmental protection, maritime safety, and national security considerations.

In practice, repair operations often require diplomatic coordination because cable systems frequently cross multiple maritime jurisdictions.

Continental Shelf

UNCLOS defines the continental shelf as the natural prolongation of a coastal state's land territory beneath the sea.

Coastal states possess **sovereign rights** over the exploration and exploitation of natural resources located on or beneath the continental shelf. These rights extend to offshore oil, gas, and mineral resources.

Importantly, however, other states retain the right to lay and maintain submarine communication cables across another state's continental shelf, provided they respect the coastal state's resource rights and environmental regulations.

Exclusive Economic Zone (EEZ)

The Exclusive Economic Zone extends **up to 200 nautical miles** from a coastal state's baseline.

Within the EEZ, coastal states enjoy sovereign rights over natural resource exploration, fisheries, offshore energy development, and environmental protection. At the same time, other states continue to enjoy internationally recognised freedoms of navigation, overflight, and the laying and maintenance of submarine cables and pipelines.

Because many cable systems traverse multiple EEZs, effective infrastructure protection often depends upon international cooperation and coordination among coastal states.

High Seas

The high seas comprise maritime areas beyond national jurisdiction.

UNCLOS guarantees all states equal freedoms on the high seas, including navigation, scientific research, fishing, overflight, and the laying of submarine cables and pipelines.

Because no single state exercises sovereignty over the high seas, responsibility for protecting critical infrastructure located there depends upon international cooperation, flag state jurisdiction, and adherence to international law.

Convention for the Protection of Submarine Telegraph Cables (1884)

The **Convention for the Protection of Submarine Telegraph Cables**, adopted in **1884**, represents the first international treaty specifically dedicated to protecting submarine communications infrastructure. Although negotiated during the era of telegraph cables, many of its principles remain relevant to modern fibre-optic cable systems.

The Convention requires states to criminalise the intentional or negligent damage of submarine telegraph cables outside territorial waters and establishes legal procedures concerning liability, repairs, and enforcement.

While technological developments have significantly transformed submarine communications, the Convention established the enduring international principle that submarine communication networks deserve special legal protection because of their global importance.

United Nations Charter

The **United Nations Charter**, adopted in **1945**, forms the constitutional foundation of the modern international legal order. Although it does not specifically address submarine infrastructure, several provisions are directly relevant when infrastructure incidents involve international disputes or the use of force.

Article 2(4)

Article 2(4) prohibits states from threatening or using force against the territorial integrity or political independence of any other state.

If deliberate attacks on critical undersea infrastructure are determined to constitute a prohibited use of force, they may engage the responsibilities of the attacking state under international law. However, determining whether sabotage, cyber operations, or hybrid attacks reach the legal threshold of "use of force" remains a subject of ongoing scholarly and governmental debate.

Article 51

Article 51 recognises the inherent right of individual and collective self-defence if an armed attack occurs against a Member State.

One of the principal legal challenges in contemporary hybrid warfare is determining whether attacks on submarine cables, offshore energy facilities, or digital infrastructure constitute an "armed attack" sufficient to justify self-defence.

This question depends upon factors such as the scale, effects, intent, and consequences of the incident, and no universally accepted legal standard currently exists.

Peaceful Settlement of Disputes

The UN Charter requires states to settle international disputes through peaceful means, including negotiation, mediation, arbitration, judicial settlement, and other peaceful procedures.

Given the frequent uncertainty surrounding attribution of undersea infrastructure incidents, peaceful dispute resolution remains particularly important for preventing escalation arising from competing accusations or incomplete investigations.

Budapest Convention on Cybercrime

The **Convention on Cybercrime**, commonly known as the **Budapest Convention**, was adopted by the Council of Europe in **2001** and remains the first and most comprehensive international treaty addressing cybercrime.

The Convention harmonises criminal offences relating to illegal access, data interference, system interference, computer-related fraud, and other cyber offences while facilitating international cooperation in cyber investigations.

Although it does not specifically regulate cyber warfare or military cyber operations, the Convention provides an important legal framework for investigating cyber incidents affecting infrastructure operated by civilian entities, including telecommunications systems and cable landing stations.

Its growing international membership has made it an important instrument for cross-border cybercrime cooperation, though not all major cyber powers are parties to the Convention.

Tallinn Manual

The **Tallinn Manual** is a non-binding academic study prepared by independent international law experts under the auspices of the NATO Cooperative Cyber Defence Centre of Excellence.

Rather than creating new law, the Manual explains how existing international law—including the UN Charter, international humanitarian law, and the law of state responsibility—may apply to cyber operations.

The Manual discusses issues including:

- cyber operations as potential uses of force,
- armed attacks in cyberspace,
- state responsibility,
- sovereignty,
- due diligence,
- countermeasures,
- attribution,
- protection of civilian infrastructure.

Although not legally binding, the Tallinn Manual has become one of the most influential references used by governments, military planners, academics, and legal practitioners when analysing cyber operations affecting critical infrastructure.

International Maritime Organization (IMO)

The **International Maritime Organization (IMO)** is the specialised agency of the United Nations responsible for improving maritime safety, security, environmental protection, and efficient international shipping.

While the IMO does not regulate submarine communication cables directly, its conventions and guidelines contribute significantly to protecting maritime infrastructure by promoting:

- safe navigation,
- vessel traffic management,
- maritime security,
- search and rescue,
- pollution prevention,
- cyber risk management for shipping,
- port security.

IMO guidance encourages shipping companies and port authorities to strengthen cybersecurity, improve operational resilience, and reduce risks that could indirectly affect critical undersea infrastructure.

International Cable Protection Committee (ICPC)

Established in **1958**, the **International Cable Protection Committee (ICPC)** is the leading international organisation dedicated to the protection of submarine communication cables.

The ICPC brings together cable owners, telecommunications companies, governments, manufacturers, research institutions, and maritime organisations to promote the security and resilience of global cable systems.

Its principal functions include:

Role

- Promoting awareness of submarine cable protection.
- Advising governments on legal and regulatory issues.
- Supporting cooperation between cable operators and maritime industries.
- Encouraging international collaboration following cable incidents.

Standards and Best Practices

The ICPC develops widely recognised industry guidance concerning:

- cable route planning,
- installation practices,
- repair procedures,

- fishing and anchoring risk reduction,
- marine spatial planning,
- environmental protection,
- emergency response,
- information sharing.

Although the ICPC possesses no enforcement authority, its technical guidance is widely adopted by governments and the submarine cable industry.

International Telecommunication Union (ITU)

The **International Telecommunication Union (ITU)** is the specialised United Nations agency responsible for global telecommunications and information and communication technologies (ICTs).

The ITU plays an important coordinating role by:

- developing international telecommunications standards,
- promoting interoperability between communication networks,
- supporting spectrum management,
- improving global digital connectivity,
- assisting developing countries in expanding communications infrastructure.

Although it does not own or regulate submarine cables directly, the ITU contributes to international cooperation concerning the resilience, standardisation, and development of global telecommunications infrastructure, including submarine communication systems.

International Seabed Authority (ISA)

The **International Seabed Authority (ISA)** was established under UNCLOS to organise and regulate mineral-related activities in the deep seabed beyond national jurisdiction, known as "**the Area.**"

The ISA's primary responsibilities include:

- regulating deep-seabed mineral exploration and exploitation,
- protecting the marine environment,
- developing environmental standards,
- issuing contracts for seabed activities,
- ensuring that activities in the Area are conducted for the benefit of humankind.

Although the ISA does not regulate submarine communication cables directly, its work is increasingly relevant because future deep-sea mining operations may interact with existing cable routes. As both activities expand, coordination between seabed resource development

and the protection of critical undersea infrastructure will become increasingly important to avoid conflicts, environmental harm, and operational interference.

Limitations of the Current Legal Framework

Despite this extensive body of international law, significant challenges remain. Most existing legal instruments were drafted before the emergence of fibre-optic communications, autonomous underwater vehicles, cyber warfare, and hybrid threats. As a result, they provide only limited guidance on issues such as cyberattacks against cable management systems, the use of autonomous underwater vehicles for sabotage, attribution of covert operations, coordinated hybrid campaigns, or thresholds for invoking self-defence.

These legal ambiguities have prompted growing calls for enhanced international cooperation, improved information sharing, updated norms of responsible state behaviour, and the development of more comprehensive legal and policy frameworks capable of addressing the evolving security challenges facing critical undersea and maritime infrastructure.

Part VI – Emerging Technologies

Introduction

Rapid technological advancement is fundamentally reshaping the protection, monitoring, and operation of critical undersea and maritime infrastructure. As hybrid threats become increasingly sophisticated, governments, private industry, and international organizations are investing in advanced technologies capable of improving surveillance, early warning, predictive maintenance, cybersecurity, and operational resilience.

Many of these technologies have dual-use characteristics, meaning they can be employed for both civilian and military purposes. While they offer significant opportunities to strengthen the security of submarine communication cables and other undersea infrastructure, they also introduce new strategic challenges relating to cybersecurity, privacy, autonomous weapons, technological competition, and international governance.

The future security of critical undersea infrastructure will therefore depend not only on traditional naval capabilities but also on advances in artificial intelligence, robotics, satellite systems, quantum technologies, and integrated digital networks.

Artificial Intelligence (AI)

Artificial Intelligence (AI) refers to computer systems capable of performing tasks that normally require human intelligence, including data analysis, pattern recognition, decision-making, and prediction. In the maritime domain, AI is increasingly used to process vast quantities of information collected from satellites, underwater sensors, sonar systems, autonomous vehicles, and communication networks.

AI enables operators to detect unusual vessel behaviour, identify anomalies in network traffic, predict equipment failures, and support faster responses to potential threats. Machine learning algorithms can continuously analyse historical and real-time data to distinguish normal operational patterns from activities that may indicate sabotage, cyber intrusions, or unauthorized underwater operations.

However, AI also presents new risks. Malicious actors may use AI to automate cyberattacks, generate convincing disinformation, enhance autonomous weapons, or exploit vulnerabilities faster than traditional security systems can respond. Consequently, AI has become both a defensive capability and a potential source of emerging hybrid threats.

Autonomous Underwater Vehicles (AUVs)

Autonomous Underwater Vehicles (AUVs) are robotic underwater platforms capable of operating independently without continuous human control. Using onboard sensors, navigation systems, and artificial intelligence, AUVs can perform complex missions over extended distances while collecting detailed information about the seabed.

Civilian applications include seabed mapping, environmental monitoring, cable inspection, offshore energy maintenance, and scientific research. Military applications include mine countermeasures, intelligence collection, anti-submarine warfare, and maritime surveillance.

As AUV technology becomes more affordable and capable, concerns have grown regarding their potential use in covert infrastructure surveillance, unauthorized seabed mapping, or deliberate interference with submarine communication cables and pipelines.

Large Underwater Drones

Large underwater drones represent the next generation of autonomous maritime systems. Compared with conventional AUVs, these vehicles possess greater endurance, increased payload capacity, and the ability to operate for weeks or even months without human intervention.

Some systems are designed to travel thousands of kilometres while carrying sophisticated sensors, communication equipment, or specialized payloads. Their long endurance enables persistent monitoring of critical infrastructure, maritime chokepoints, and underwater environments.

While these capabilities improve maritime security, they also create concerns regarding covert surveillance, underwater sabotage, and strategic competition, particularly as several major powers continue to expand autonomous naval capabilities.

Deep-Sea Robotics

Deep-sea robotics encompasses remotely operated vehicles (ROVs), autonomous systems, robotic manipulators, and specialised engineering equipment capable of operating at extreme ocean depths.

These systems play an essential role in inspecting, repairing, and maintaining submarine cables, offshore energy platforms, pipelines, and scientific observatories. They can access environments that are inaccessible or hazardous for human divers.

Recent advances in robotics have improved precision, endurance, and automation. Nevertheless, the same technologies could potentially be adapted for unauthorized inspection, tampering, or sabotage of undersea infrastructure, highlighting the dual-use nature of deep-sea robotics.

Quantum Communication

Quantum communication is an emerging field that uses the principles of quantum mechanics to transmit information with exceptionally high levels of security. Unlike conventional communication systems, quantum communication relies on quantum states that cannot be measured or copied without altering them.

This characteristic offers the potential to detect any attempt at interception, making quantum communication particularly attractive for diplomatic, financial, and military communications transmitted through future undersea networks.

Although widespread deployment remains limited, continued research suggests that quantum communication may significantly enhance the long-term security of global communications infrastructure.

Quantum Encryption

Quantum encryption, particularly **Quantum Key Distribution (QKD)**, enables two parties to exchange cryptographic keys using quantum particles such as photons. Because any attempt to intercept the transmission changes the quantum state of the particles, eavesdropping becomes immediately detectable.

Future submarine communication cables may increasingly incorporate quantum communication technologies to protect sensitive government, financial, and defence communications against both current and future cyber threats, including attacks from quantum computers.

At the same time, the emergence of quantum computing may eventually challenge many existing encryption methods, requiring substantial upgrades to global communications infrastructure.

Satellite Integration

Although submarine cables carry the overwhelming majority of global data traffic, satellites provide essential complementary capabilities. Modern infrastructure increasingly integrates satellite communications with terrestrial and submarine networks to improve redundancy, resilience, and operational continuity.

Satellite systems support:

- navigation,
- positioning,
- timing,
- emergency communications,
- infrastructure monitoring,
- maritime surveillance,
- disaster response.

If submarine cable systems are disrupted, satellite communications can provide temporary backup connectivity for critical government and emergency services, although they generally cannot replace the full capacity of fibre-optic networks.

Space-Based Monitoring

Earth observation satellites increasingly contribute to the protection of maritime infrastructure by providing continuous surveillance of the oceans.

Using optical imagery, synthetic aperture radar (SAR), thermal sensors, and radio-frequency detection systems, satellites can monitor:

- vessel movements,
- suspicious maritime activity,
- illegal fishing,
- oil spills,
- offshore energy installations,
- maritime traffic patterns.

Satellite monitoring significantly improves Maritime Domain Awareness (MDA) by enabling authorities to identify abnormal behaviour near critical infrastructure, even in remote ocean regions.

Digital Twins

A digital twin is a virtual representation of a physical asset that continuously updates using real-time operational data collected from sensors and monitoring systems.

For submarine communication cables, offshore platforms, or pipelines, digital twins enable operators to simulate equipment performance, monitor infrastructure health, predict maintenance requirements, and evaluate responses to potential failures or attacks.

By modelling the entire lifecycle of critical infrastructure, digital twins improve operational efficiency while reducing maintenance costs and enhancing resilience against unexpected disruptions.

Machine Learning for Anomaly Detection

Machine learning is a branch of artificial intelligence that enables computer systems to recognise patterns and improve performance through experience rather than explicit programming.

In critical infrastructure protection, machine learning algorithms analyse enormous volumes of operational data to identify abnormal patterns that may indicate equipment failures, cyber intrusions, unauthorized vessel activity, or sabotage.

For example, algorithms may detect unusual fluctuations in cable performance, unexpected vessel movements near cable routes, or suspicious network behaviour long before human operators recognise a developing incident.

Such predictive capabilities enable earlier intervention and reduce the likelihood of major infrastructure failures.

Acoustic Sensing

Acoustic sensing exploits the efficient transmission of sound through seawater to detect underwater activity.

Hydrophones and advanced acoustic sensors monitor sounds generated by ships, submarines, underwater drones, marine mammals, geological activity, and human operations on the seabed.

Modern acoustic sensing systems increasingly incorporate artificial intelligence to distinguish between normal environmental sounds and potentially suspicious activity, improving underwater situational awareness.

Acoustic monitoring forms a critical component of both civilian infrastructure protection and military anti-submarine operations.

Seabed Sensors

Seabed sensors are permanently installed monitoring devices placed directly on or beneath the ocean floor.

These sensors may measure:

- pressure,
- temperature,
- seismic activity,
- acoustic signals,
- magnetic fields,
- ocean currents,
- equipment integrity.

Networks of seabed sensors can provide continuous monitoring of submarine communication cables, pipelines, offshore energy infrastructure, and strategic maritime areas.

Integrated sensor networks improve early warning capabilities by rapidly detecting unauthorized underwater activity or changes in infrastructure condition.

Smart Cables

Smart cables are next-generation submarine communication cables that combine traditional fibre-optic communications with integrated environmental sensing technologies.

In addition to transmitting internet traffic, smart cables may continuously measure:

- water temperature,
- pressure,
- seismic activity,
- ocean currents,
- tectonic movement.

This dual functionality allows communication infrastructure to contribute simultaneously to scientific research, earthquake detection, tsunami warning systems, and climate monitoring without requiring separate sensor networks.

As smart cable technology develops, these systems are expected to become important components of both digital infrastructure and global environmental observation networks.

Ocean Internet of Things (Ocean IoT)

The Ocean Internet of Things (Ocean IoT) refers to interconnected networks of underwater sensors, autonomous vehicles, offshore platforms, buoys, communication systems, and environmental monitoring devices that exchange information in real time.

Similar to terrestrial IoT systems, Ocean IoT enables continuous monitoring of distributed infrastructure through intelligent devices connected by communication networks.

Potential applications include:

- infrastructure monitoring,
- offshore energy management,
- maritime traffic surveillance,
- environmental protection,
- fisheries management,
- disaster warning,
- scientific research.

However, the increasing connectivity of underwater infrastructure also expands the cyber attack surface. Poorly secured sensors or communication devices may become entry points for cyber intrusions, making cybersecurity an essential component of future Ocean IoT development.

The Future of Undersea Infrastructure Security

Emerging technologies are transforming undersea infrastructure from static physical assets into intelligent, interconnected systems capable of monitoring their own condition, detecting threats, and supporting rapid decision-making. Artificial intelligence, autonomous systems, satellite integration, advanced sensing technologies, and digital modelling offer unprecedented opportunities to enhance resilience against hybrid threats.

At the same time, these technologies introduce new legal, ethical, and security challenges. Their dual-use nature means that innovations designed to improve inspection, maintenance, and scientific research may also be exploited for surveillance, cyber operations, or sabotage. Consequently, future governance of critical undersea and maritime infrastructure will require not only technological innovation but also international cooperation, common standards, cybersecurity safeguards, and the development of norms governing the responsible use of emerging technologies in the maritime domain.

Major Case Studies

Introduction

Although submarine communication cables and other forms of critical undersea infrastructure have operated for over a century, recent years have witnessed a growing number of incidents involving physical damage, suspected sabotage, cyber concerns, and geopolitical tensions. While some incidents resulted from natural disasters or accidental human activity, others occurred amid periods of heightened international rivalry, making attribution difficult and raising concerns regarding hybrid warfare.

These case studies demonstrate the diverse threats facing undersea infrastructure—from earthquakes and volcanic eruptions to covert intelligence operations, military competition, and grey-zone tactics. They also highlight recurring challenges concerning attribution, legal responsibility, infrastructure resilience, and international cooperation.

Nord Stream Pipeline Explosions (2022)

Background

On **26 September 2022**, powerful underwater explosions damaged the **Nord Stream 1** and **Nord Stream 2** natural gas pipelines beneath the Baltic Sea. The pipelines, constructed to

transport Russian natural gas directly to Germany, represented one of Europe's most strategically important energy infrastructure projects.

The explosions occurred during heightened geopolitical tensions following Russia's invasion of Ukraine, immediately transforming the incident into one of the most significant infrastructure security events in recent history.

Why It Is Important

The Nord Stream incident fundamentally changed international perceptions regarding the vulnerability of undersea infrastructure. It demonstrated that offshore energy assets located far beneath the sea could become targets during periods of geopolitical confrontation.

Beyond the immediate loss of gas transportation capacity, the incident affected European energy markets, intensified diplomatic tensions, and prompted governments to reassess the protection of submarine cables, pipelines, and offshore installations.

The event also accelerated NATO and European Union initiatives concerning critical infrastructure security.

Hybrid Warfare Significance

The explosions illustrated many characteristics commonly associated with hybrid warfare.

The attack targeted civilian economic infrastructure rather than conventional military objectives.

It occurred outside active battlefields.

Responsibility remained uncertain.

The incident generated extensive political debate, information campaigns, intelligence activity, and competing public narratives.

Regardless of attribution, the event demonstrated how attacks on critical infrastructure could produce strategic effects without involving conventional military confrontation.

Investigation Challenges

Investigating underwater infrastructure incidents presents exceptional technical difficulties.

Evidence is located beneath the sea.

Environmental conditions rapidly degrade forensic material.

Multiple jurisdictions may be involved.

Access requires specialised underwater equipment.

Consequently, investigations often require months or years while involving police agencies, intelligence services, naval authorities, and international cooperation.

Attribution Issues

One of the most significant challenges following the Nord Stream explosions concerned attribution.

Various investigations have explored multiple hypotheses, while responsibility remains politically contested. The absence of definitive public attribution illustrates one of the central characteristics of hybrid threats: uncertainty regarding the identity of the responsible actor can itself become a strategic advantage.

Balticconnector Pipeline Incident (2023)

Background

In October 2023, the **Balticconnector** natural gas pipeline linking Finland and Estonia was damaged in the Gulf of Finland, leading to an interruption of gas flows. A nearby telecommunications cable was also damaged during the incident.

Subsequent investigations suggested that the damage was likely caused by a ship's anchor dragging across the seabed, though authorities continued examining the circumstances surrounding the event.

Why It Matters

The incident demonstrated how a single seabed event could simultaneously affect both energy infrastructure and communication systems.

It also highlighted the vulnerability of relatively shallow seabed infrastructure located within heavily trafficked shipping routes.

Anchor Drag Investigation

Investigators analysed vessel movements, recovered damaged infrastructure, examined anchor marks on the seabed, and reviewed Automatic Identification System (AIS) data.

The investigation illustrated the complexity of distinguishing accidental damage from deliberate interference, particularly when commercial vessels operate near critical infrastructure.

Baltic Sea Communication Cable Incidents (2023–2025)

Background

Between 2023 and 2025, several submarine communication cables in the Baltic Sea experienced unexpected damage. Some incidents occurred alongside disruptions affecting pipelines or power cables.

Several investigations examined whether vessel anchor dragging, inadequate seamanship, or deliberate actions contributed to the damage.

Suspicious Vessel Movements

Authorities increasingly relied upon satellite imagery, AIS tracking, radar surveillance, and underwater inspections to reconstruct vessel movements before each incident.

Repeated attention focused on commercial ships whose movements appeared unusual near cable routes, illustrating how hybrid threats may exploit civilian maritime activity.

Lessons Learned

The Baltic incidents demonstrated the need for:

- improved maritime surveillance,
- stronger cooperation among neighbouring states,
- faster information sharing,
- enhanced repair capabilities,
- greater redundancy within communication networks,
- better protection of critical infrastructure during peacetime.

Svalbard Cable Incidents

Background

The Norwegian archipelago of **Svalbard** hosts strategically important submarine communication cables connecting Arctic satellite ground stations with mainland Norway.

Past cable failures disrupted scientific data transmission and communications supporting Arctic research.

Importance

Svalbard occupies a strategically significant location within the Arctic, supporting satellite operations, scientific research, and growing polar maritime activity.

Its communication infrastructure is therefore increasingly important for both civilian and strategic purposes.

Arctic Implications

The incidents highlighted the challenges of protecting infrastructure in remote Arctic waters where severe weather, ice conditions, and limited repair capabilities complicate response efforts.

As Arctic maritime activity expands, infrastructure resilience is expected to become an increasingly important policy issue.

Shetland Islands Cable Disruption

The Shetland Islands rely heavily upon submarine communication cables connecting them with mainland Scotland.

Past cable failures temporarily disrupted internet services, telephone communications, banking operations, and emergency communications across the islands.

Although caused by infrastructure failures rather than deliberate attacks, the incident demonstrated how communities dependent upon limited communication links remain particularly vulnerable.

It also reinforced the importance of redundancy and diversified communication networks.

Matsu Islands Cable Cuts (Taiwan)

Background

Taiwan's **Matsu Islands**, located close to mainland China, have experienced repeated submarine cable disruptions affecting internet connectivity.

These incidents significantly affected communications for local residents and government services.

Chinese Vessel Concerns

Some investigations examined whether vessels operating near cable routes contributed to the damage.

Because the incidents occurred amid broader cross-strait tensions, they attracted considerable international attention despite the absence of definitive evidence regarding deliberate sabotage.

Communications Disruption

The cable cuts reduced internet speeds, interrupted digital services, and required temporary reliance on satellite communications.

The incidents demonstrated the vulnerability of geographically isolated communities dependent upon only a few communication links.

Resilience Measures

Taiwan subsequently strengthened communication resilience through:

- additional cable redundancy,
- expanded satellite backup,
- improved repair planning,
- enhanced monitoring of cable routes.

These measures are increasingly cited as examples of infrastructure resilience planning.

Red Sea Cable Damage (2024)

Background

During 2024, several submarine communication cables in the Red Sea suffered damage amid escalating regional instability.

Because the Red Sea contains one of the world's most concentrated cable corridors, disruptions affected internet connectivity across Europe, Asia, Africa, and the Middle East.

Conflict Spillover

Although submarine cables were not necessarily the primary targets, the incidents demonstrated how regional armed conflicts may indirectly threaten civilian communication infrastructure.

Operations affecting shipping, military activity, and offshore security created additional risks for repair operations.

Regional Internet Disruption

Damage to multiple cables reduced communication capacity, increased internet latency, and required traffic to be rerouted through alternative international cable systems.

The incidents highlighted the importance of network redundancy in maintaining global connectivity.

Yemen Conflict

The prolonged conflict in Yemen has transformed the southern Red Sea into a strategically sensitive maritime environment.

Military operations, naval deployments, mines, missile attacks, and security risks have complicated commercial shipping and increased concerns regarding nearby submarine communication cables and offshore infrastructure.

The conflict demonstrates how instability on land can generate significant risks for maritime infrastructure beyond the immediate conflict zone.

Houthi Attacks

Beginning in late 2023, Houthi forces launched numerous attacks against commercial shipping transiting the Red Sea.

Although these attacks primarily targeted merchant vessels, they significantly affected maritime traffic patterns.

Shipping companies increasingly diverted vessels around the Cape of Good Hope, increasing transit times and costs.

The concentration of military activity, damaged vessels, and security operations also increased indirect risks to submarine cable maintenance and repair operations in the region.

Russian Naval Activity Around Submarine Cables

Background

Several NATO governments have expressed concern regarding increased Russian naval activity near important submarine communication cable routes in the North Atlantic and surrounding European waters.

Although operating in international waters is lawful, repeated patrols near critical infrastructure have heightened strategic attention.

Yantar

The Russian oceanographic research vessel **Yantar** has attracted particular attention because it possesses sophisticated deep-sea submersibles capable of operating near submarine cables and seabed infrastructure.

Russian authorities describe the vessel as supporting oceanographic research.

Western security analysts have suggested that its specialised underwater capabilities could also support intelligence collection or seabed operations, although public evidence regarding specific activities remains limited.

Specialized Underwater Capabilities

Modern naval powers increasingly operate:

- deep-diving submersibles,
- autonomous underwater vehicles,
- seabed mapping systems,
- underwater robotics.

These capabilities support legitimate scientific and military missions but also illustrate how technological advances may facilitate covert operations affecting critical infrastructure.

Cold War Cable Espionage

Throughout the Cold War, submarine communication cables became valuable intelligence targets.

Both the United States and the Soviet Union sought access to underwater communications to improve strategic intelligence, monitor military activity, and support national security planning.

These operations demonstrated that undersea infrastructure has long possessed strategic importance extending beyond commercial communications.

Operation Ivy Bells

Operation **Ivy Bells** remains one of the most famous undersea intelligence operations ever conducted.

Beginning in the early 1970s, the United States deployed specially equipped submarines and divers to secretly access Soviet underwater military communication cables in the Sea of Okhotsk.

Rather than cutting the cables, intelligence personnel installed sophisticated recording devices capable of intercepting communications without immediately alerting Soviet authorities.

The operation continued for several years before eventually being compromised.

Ivy Bells demonstrated the strategic intelligence value of submarine communication infrastructure and influenced subsequent developments in cable security and encryption.

NSA Undersea Interception Programs

Historical disclosures indicate that intelligence agencies have long regarded submarine communication systems as strategically important sources of information.

Over time, advances in fibre-optic encryption have made direct interception significantly more difficult than during earlier decades.

Nevertheless, concerns regarding lawful interception, metadata collection, infrastructure access, and communications security continue to influence debates regarding submarine cable governance and cybersecurity.

Earthquake-Induced Cable Failures

Japan

Japan experiences frequent seismic activity, making submarine communication cables vulnerable to earthquakes and associated underwater landslides.

Several major earthquakes have damaged multiple cable systems simultaneously, temporarily reducing international communication capacity.

Taiwan

Taiwan has similarly experienced repeated cable disruptions resulting from earthquakes.

These incidents demonstrated that natural hazards may produce consequences comparable to deliberate attacks by affecting multiple communication routes simultaneously.

Lessons on Resilience

Earthquake-related failures have encouraged cable operators to:

- diversify cable routes,
- improve network redundancy,
- strengthen disaster planning,
- accelerate repair capabilities,
- develop more resilient infrastructure designs.

Tonga Volcanic Eruption (2022)

In January 2022, the eruption of the **Hunga Tonga–Hunga Ha'apai** underwater volcano severed Tonga's primary submarine communication cable.

The country experienced widespread communication outages, significantly affecting emergency coordination, financial services, humanitarian assistance, and contact with the international community.

Because alternative communication systems were limited, restoration required specialised repair vessels travelling long distances.

The Tonga case highlighted how natural disasters can isolate entire nations and demonstrated the importance of resilient communication infrastructure, backup connectivity, and rapid international assistance mechanisms.

Fukushima and Infrastructure Resilience

The **2011 Great East Japan Earthquake and Tsunami**, which triggered the Fukushima Daiichi nuclear accident, also affected a wide range of coastal infrastructure, including ports, energy systems, telecommunications, and transportation networks.

Although submarine communication cables proved comparatively resilient, the broader disaster demonstrated that critical infrastructure systems are highly interconnected. Damage to electricity, transport, logistics, and emergency communications can significantly delay infrastructure recovery even when individual cable systems remain operational.

The Fukushima experience reinforced the importance of integrated resilience planning, redundancy across critical infrastructure sectors, robust emergency response mechanisms,

and disaster-resilient infrastructure design—principles that remain highly relevant to protecting modern undersea and maritime infrastructure.

Questions A Resolution Must Answer (QARMA)

- Should attacks on submarine cables be explicitly classified as threats to international peace and security?
- How can attribution standards for hybrid maritime attacks be strengthened?
- Should the UN establish an international mechanism for monitoring critical undersea infrastructure?
- What responsibilities should private cable owners bear for resilience and cybersecurity?
- Should states permit international inspections of underwater infrastructure in their EEZs?
- How can freedom of navigation be balanced with protection of critical infrastructure?
- What role should autonomous underwater systems play in infrastructure surveillance?
- How can developing states improve cable resilience despite limited resources?
- Should there be mandatory information-sharing following cable disruptions?
- How can international law address grey-zone activities below the threshold of armed conflict?

CONCLUSION

Good Luck Delegates. May the best delegate emerge with the best solutions, better lobby, better questions and better documentation :)

Note: No position papers will be accepted.

No part of this draft can be used in the committee as an official source of information.